

Progress in Anomalous Transport Research in Toroidal Magnetic Confinement Devices

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Abstract—Transport is the outstanding physics issue in the quest for fusion by magnetic confinement. In spite of the intrinsic difficulty, a great deal of progress has been made in the past 25 years. Experiments have gone from being dominated by high anomalous losses, of the order of Bohm diffusion losses, to operation with no anomalous transport. This success is due to a combination of improved experimental infrastructure and the high degree of knowledge on how to control plasma discharges. Both have made it possible to access enhanced confinement regimes and to unravel new effects in confinement physics. Although there is not yet a complete understanding of the dynamical mechanisms underlying the anomalous transport process, there is some understanding of important components such as the ion transport loss mechanism at the plasma core and of the main mechanism for turbulence suppression in the enhanced confinement regimes.

Index Terms—Plasma confinement, plasma devices, plasma heating, plasma measurements, plasma stability.

I. INTRODUCTION

TRANSPORT is the outstanding critical physics issue in fusion by magnetic confinement. Its direct impact on the size of the reactor and thereby cost and practicality make it a critical issue. Its intrinsic and fundamental scientific complexity has not yet allowed its complete resolution. The theoretical bases for the classical transport in toroidal confinement systems (neoclassical theory) [1], [2] are well established. The problem is that, for the usual low confinement and ohmic regimes in tokamaks, the neoclassical transport theory predicts electron and ion confinement times up to two orders of magnitude longer than those experimentally measured. Plasma turbulence is probably the major cause of plasma cross-field transport and the mechanism for this so-called anomalous transport (nonclassical transport). Therefore, the real problem of understanding anomalous transport is understanding plasma turbulence. The scientific difficulties of understanding turbulence in classical fluids are well known [3]. In magnetically confined plasmas, the problem of turbulence is even more complex. The problem involves multiple charged turbulent fluids coupled through an electromagnetic field. In experiments, apart from the plasma physics transport processes, plasma confinement is strongly affected by edge physics and wall conditions. The connection between these

effects and plasma core confinement is not yet well understood. This coupling between plasma transport and edge physics effects increases the difficulty in interpreting the experimental results.

In spite of these intrinsic difficulties, a great deal of progress has been made. This progress can be viewed as progress in understanding and in controlling the negative effects of anomalous transport on confinement. Progress on the latter can be clearly measured by comparing the aim of the research 25 years ago with present results. At that time, this goal was clearly stated by Kadomtsev and Pogutse [4]: “The question of achieving controlled fusion thus reduces to the possibility of reducing the turbulent diffusion coefficient to a value that is two orders of magnitude smaller than the Bohm value.” The Bohm diffusion coefficient is $D_{\text{Bohm}} = cT/16|e|B$. The way this goal was stated can now be seen as somewhat narrow, but it illustrates the perspective of the time. Recent experiments in TFTR [5] and DIII-D [6], [7] tokamaks have shown that in the region of negative magnetic shear, transport drops by nearly two orders of magnitude to the neoclassical level. This operational regime has led to the achievement of an effective break-even in JT-60U [8]. These transport levels verify well the conditions of [4] with turbulent transport effectively eliminated at the plasma core. Therefore, the program has succeeded well beyond expectations. Of course, these experimental results that show total suppression of the anomalous transport hold only in a transient way. However, we expect that sustainment over long pulses will be demonstrated in the near future. If that is the case, transport by itself will no longer be a practical issue impeding the achievement of controlled nuclear fusion. The issue becomes the control and sustainment of the plasma conditions for these advanced operational regimes. What also remains is the need for full understanding of the transport mechanisms, which will enable us to optimize fusion reactor performance. Of course, understanding turbulent transport in plasmas remains a scientific issue by itself and for its importance in other research areas where plasma physics plays an essential role.

To reach the successful stage in confinement performance, we have gone through some ups and downs. The 1970's began with the optimism of the good confinement in T-3 tokamak making the tokamak the lead confinement concept. At this time, there were also some concerns about the effect of trapped particle modes on confinement of high-temperature plasmas. When temperatures of 20 keV were reached in PLT, some of these fears dissipated. A few years later, as the auxiliary heating in tokamaks increased well above the ohmic

Manuscript received December 11, 1996; revised July 31, 1997. This work was sponsored by the Office of Fusion Energy, U.S. Department of Energy, under Contract DE-AC05-96OR22464 with Lockheed Martin Energy Research Corp.

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Publisher Item Identifier S 0093-3813(97)08989-3.

heating, this optimistic result was followed by the realization that confinement deteriorated with increasing power. This confinement regime has been called low-confinement mode (L-mode). The reason for this name is that, in the early 1980's, a new confinement regime was found for which confinement time is about twice the L-mode confinement. The new confinement regime was called high-confinement mode (H-mode). To access this mode, high power (above a threshold value) and careful wall conditioning are needed. Finding the H-mode was only the beginning. From the mid 1980's to the 1990's, a large number of enhanced confinement modes have been found with better and better confinement. This process has culminated with the aforementioned recent results for which anomalous transport is practically eliminated. This spectacular progress in the last decade has required high-power heating, sophisticated wall-conditioning techniques, and a high degree of control of the density, temperature, and safety factor q profiles. It has involved a learning process of controlling the plasma discharges and understanding how to optimize them.

For many years, theory has identified the possible mechanisms of turbulence in magnetically confined plasmas (instability drives and linear instability theory). A major problem is the difficulty of defining a direct yes or no experimental test of those basic mechanisms. The complexity of the plasma physics experiments and the strong nonlinear couplings between the observables makes it very difficult and sometimes serves to mislead the experimental tests of theory. The challenge is in constructing a transport theory from the fundamental mechanisms and in formulating the right experimental tests. Steady progress has been made in this direction; an example is the understanding of anomalous transport in the ion channel at the plasma core. New powerful concepts have also been developed as a response to recent experimental progress. Some of the more successful ones are the idea of $E \times B$ shear suppression of turbulence and the beta-induced stabilization of curvature-driven modes (second stability).

It is useful to formulate the transport problem in a macroscopic way. On transport time scales, a fluid representation can be used [9]. For axisymmetric configurations, a one-dimensional representation can be obtained by averaging plasma observables on a flux surface over the poloidal and toroidal angles. In this way a set of one-dimensional fluid-like equations can be derived to represent the evolution of the density n_a , momentum $m_a n_a \vec{V}_a$, and energy $n_a T_a$ for each plasma species

$$\frac{\partial n_a}{\partial t} = -\frac{1}{r} \frac{\partial}{\partial r} (r \Gamma_{pa}) + S_a \quad (1)$$

$$m_a n_a \frac{\partial V_{ia}}{\partial t} = -\frac{1}{r} \frac{\partial}{\partial r} [r \Phi_{ia}] + M_{ia} \quad (2)$$

$$\frac{3}{2} n_a \frac{\partial T_a}{\partial t} = -\frac{1}{r} \frac{\partial}{\partial r} \left[r \left(Q_a + \frac{5}{2} \Gamma_{pa} T_a \right) \right] + P_a \quad (3)$$

and the poloidal magnetic field B_p diffusion equation

$$\frac{\partial B_p}{\partial t} = \frac{\partial}{\partial r} \left[\frac{\eta}{\mu_0 r} \frac{\partial (r B_p)}{\partial r} \right] - \frac{\partial}{\partial r} (\eta J_b). \quad (4)$$

Here, the subindex a indicates the particle species, the subindex i indicates the components of the velocity vectors,

Γ_{pa} , Φ_{ia} , and Q_a are the particle, momentum, and energy fluxes, and S_a , M_{ia} , and P_a are the source, sink, and transfer terms for particles, momentum, and energy. In the magnetic field diffusion equation (4), η is the resistivity and J_b is the bootstrap current [10]. The radius r is a generalized minor radius that labels each magnetic surface. In the second-order operators, metric elements must be included to provide the information on magnetic geometry of the configuration. Here, they have not been explicitly included for simplicity. For a more detailed discussion of the transport equations, see [11]–[13].

The fluxes contain all the dynamical information on the transport processes. While neoclassical theory gives an explicit expression for them, there is not a general form for turbulent transport. In practice, these fluxes have been parameterized by analogy to the structure given by neoclassical theory. For instance, the simplest assumption of diffusive fluxes gives

$$\Gamma_{pa} = -D_a \frac{\partial n_a}{\partial r} \quad (5)$$

$$\Phi_{ia} = -\chi_{ia} \frac{\partial V_{ia}}{\partial r} \quad (6)$$

$$Q_a = -\chi_a \frac{\partial T_a}{\partial r}. \quad (7)$$

In practice, each of the fluxes depends on all the thermodynamic forces (profile gradients) and a full matrix of fluxes must be considered, including nondiffusive and nondiagonal terms.

To fully solve the transport problem (experimentally and theoretically), it is necessary to know and control:

- sources and sinks—that is, heating, fueling, current drive, and momentum sources, as well as direct particle loss, charge exchange losses, etc.;
- flux surface geometry. This requires equilibrium reconstruction that implies measuring the pressure and current profiles, and incorporating the geometry of the plasma cross-section shape;
- diffusion coefficients. The diffusion coefficient is where the microscopic dynamical information is contained. Theoretical information on the dynamics and an understanding of the plasma fluctuation is required;
- boundary conditions. We need to know not only the edge configuration, but edge physics effects and the impact of wall conditioning;
- the structure of the transport equation. Transient transport techniques are essential in probing the form of the different terms in (5)–(7).

On the first two topics listed, it is possible to obtain reliable experimental information and a good theoretical basis. The major difficulty is in dealing with the remaining three topics on the list.

Knowledge of the diffusion coefficients requires knowledge of the turbulence dynamics; this knowledge is limited. In recent years, analytical theory and nonlinear gyrofluid and gyrokinetic calculations have advanced significantly, as will be discussed in Section V. They are providing some basis for understanding the anomalous diffusivities. We can also determine these transport coefficients from experiment and we can hope for an empirical characterization of their main

parameter dependences. Even when an empirical approach is followed, these transport coefficients cannot be arbitrarily chosen. They are constrained to depend on the relevant dimensionless parameters. A great deal of research is dedicated to these topics, as we will discuss in following sections.

Our knowledge is limited regarding the boundary conditions. Boundary regions on magnetic confinement experiments are complex. They have complicated three-dimensional (3-D) geometry (divertor configurations, stochastic limiters, etc.). Atomic physics plays an essential role (interaction with neutrals and impurities) in the edge region. There is also the issue of wall conditioning, the impact of which on core physics is very important from an experimental perspective and theoretical knowledge is problematic at best. Modeling of the edge region has attained some prominence in recent years, but there is not yet enough coupling between the edge and core plasma models.

Finally, the use of perturbative experiments is very important in clarifying the structure of the anomalous fluxes and in finding some of the parametric dependences of the diffusivities. For instance, the evolution of a small-density perturbation ($\delta n/n \ll 1$) tests the diffusive nature of (1). The form of the flux Γ_α could be (and probably is) more complicated than is given by (5) and it could include terms of the form vn , a pinch term, where v is an inward velocity. Equations (1)–(4) could even include nonlocal transport terms. Also, dependence of diffusivities on density, temperature, and their gradients is tested by the perturbative experiments. Not enough perturbative experiments have been done yet, and the few existing results have raised serious questions on the traditional approach to transport. We will discuss some of these results in Section VI.

In this paper, I give an overview of the progress made during the last 25 years; it is not a comprehensive review on transport. There are excellent reviews on neoclassical transport theory [14], [15], and, the discussion herein is limited to the anomalous transport research. For a more detailed review of anomalous transport, the reader is directed to several existing review papers that cover this time period [16]–[30].

The progress of confinement studies during this 25-year period can be gauged by first identifying the status of transport studies in the early 1970's, as is done in Section II. The development in infrastructure of experiments (heating, current drive, fueling, exhaust, and diagnostics) as well as computer capabilities have been a driver for the progress in performance and understanding of confinement. A short summary of the evaluation of this infrastructure is presented in Section III, with emphasis on their impact on transport studies. Having evaluated the role of infrastructure, it is possible for us to analyze the progress made in those studies. As already mentioned, there is an obvious increase in confinement performance in the last 25 years, but there is also a corresponding progress in the experimental characterization of transport, both at macroscopic level and from the perspective of the underlying turbulence. These experimental aspects of the experimental research are presented in Section VI. Anomalous transport studies have gone to a deeper level than empirical characterization; in some areas, there is an emerging understanding. As an illus-

tration, I have selected several of these areas in Section VII for discussion. In other areas, however, there is conflicting information, and even the bases for the present understanding are being challenged; some of these open issues are presented in Section VII. With these issues in mind, a future outlook for anomalous transport research is given in Section VIII.

II. TRANSPORT UNDERSTANDING IN THE EARLY 1970'S

To better view the progress on transport studies, let us first consider the status of transport understanding at the beginning of the 1970's. Tokamaks had just been established as leading candidates for the fusion reactor when T-3 tokamak (15-cm minor radius, 1.7–3.4 T, and up to 120-kA toroidal current) showed confinement times up to 30 times the Bohm value [31]. Many other types of magnetic confinement devices were in operation at that time (stellarators, mirrors, multipoles, etc.). Let us summarily review the status of the transport research.

A. Experimental Infrastructure for Transport Studies

Heating and fueling systems were limited to plasma production. No significant auxiliary heating or alternative fueling systems were yet available, although many of the ideas for the development of those systems had already been put forward.

We can get a sense of the available diagnostic systems by looking at the description of the determination of the confinement parameters in the T-3 tokamak [31]. The averaged plasma temperature was determined from the diamagnetic loop measurement and the measurement of the plasma density with interferometry. The result for the temperature was checked by analysis of the soft X-ray spectrum. The ion temperature was determined from the energy spectrum of charge-exchanged atoms. The electron temperature measurements were confirmed later on by Thomson scattering [32]. In the early experiments, the determination of the electron temperature profile was made by obtaining a single radial point in different plasma discharges.

Fluctuation measurements by probes and their interpretation were well developed. In Zeta plasmas, density, temperature, their gradients, and the electric field fluctuations had been measured with double-Langmuir probes [33]. Mirnov loops to measure edge magnetic fluctuations were used to identify large-scale coherent MHD modes [34], and microwave scattering techniques had been proposed for density fluctuation measurements.

B. Experimental Transport Studies

The highest performance had been obtained in the aforementioned T-3 experiment where the confinement time had broken away from the usual Bohm values and reached a maximum value of 10 ms.

The electron temperature was measured with Thomson scattering with enough accuracy to show that the electron transport coefficients were close to two orders of magnitude above neoclassical values. Ion transport was assumed to be neoclassical, no profile measurements were available, and the results of modeling confinement experiments were consistent with ion neoclassical transport, often with a multiplier of the

order of two. Of course, most of these experiments were ohmically heated and there were very few experiments with auxiliary heating. The heating power level for these experiments was comparable to ohmic heating power; therefore, the separation between the electron and ion channels was difficult.

The experimental research was concerned with the determination of empirical formulas for plasma temperature and global confinement time. For instance, the data from the T-3 and TM-3 devices [31] led to

$$\tau_E \propto a^2 n_e \bar{T}^{3/2} \quad (8)$$

where

τ_E energy confinement time;

a plasma minor radius;

n_e electron density;

$\bar{T}$ averaged plasma temperature over the plasma cross section.

The classical nature of the parallel transport mechanisms was not yet confirmed, and experimental analysis was carried out to determine whether the resistivity was classical or anomalous. The relatively high impurity content of these early experimental plasmas and the relatively small number of diagnostics made this analysis still ambiguous.

Stellarator confinement was poor. In some devices, magnetic field errors were a major cause of the confinement losses. The use of ohmic heating added further complication to configurations that were supposed to operate with zero-net current. This poor performance was the main reason for the fusion program to turn to tokamaks.

C. Understanding of Anomalous Transport

At this time, the basis for the neoclassical transport theory was set, although detailed determination of the transport coefficients for realistic aspect ratios and the interpolation between different collisionality regimes were in progress. Some of the theoretical bases of radial electric field effects on particle confinement were being developed [35], [36]. These studies developed slowly for mirrors and stellarators, but would not be taken up by the tokamak until the late 1980's.

Most of the plasma instabilities had already been identified and linear theory developed. However, neither the diagnostics nor the nonlinear understanding to do detailed experimental tests of theory were yet available. Theoretical research on anomalous transport was focused mostly on trapped-particle instabilities [37], [38]. They were expected to be a threat to high-temperature operation and it was a serious source of concern.

In the following three sections of this paper, we will examine the progress made in each of the three areas of transport research just described.

III. EVOLUTION OF THE INFRASTRUCTURE FOR TRANSPORT STUDIES

In fusion research, a measure of performance is the value of the fusion triple product $n_i T_i \tau_E$, where n_i and T_i are the ion density and temperature and τ_E is the global energy confinement time. Progress made on performance in

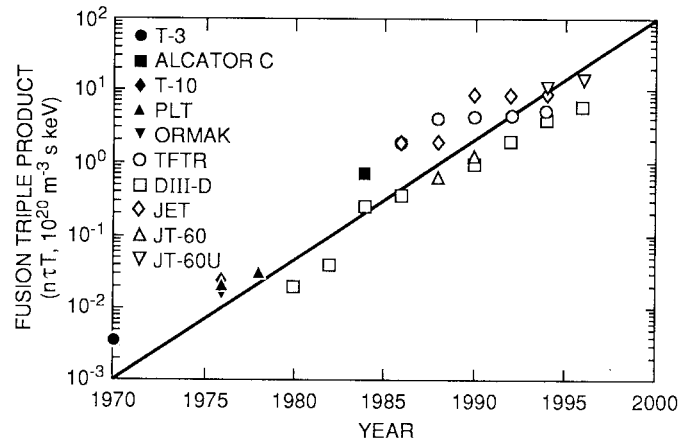


Fig. 1. The $n_i T_i \tau_E$ best values as quoted by experimental groups at the biannual International Conference Plasma Physics and Controlled Nuclear Fusion Research.

fusion experiments can be illustrated by plotting the fusion triple product versus time, as shown in Fig. 1. The $n_i T_i \tau_E$ values plotted in Fig. 1 are the best values quoted by experimental groups at the biannual International Conference on Plasma Physics and Controlled Nuclear Fusion Research. This plot shows a spectacular increase by nearly four orders of magnitude during the last 25 years. However, the way the performance measure is presented may be misinterpreted. One could read Fig. 1 to mean that by building bigger experimental facilities, we have larger values of $n_i T_i \tau_E$. Although this interpretation is valid, this statement can give a misleading interpretation of the progress in confinement and obscures one of the important roots of the progress made. It is therefore more meaningful to show the time evolution of $n_i T_i \tau_E$ in a single device. For instance, Fig. 2 shows the evolution of the best performance discharges in DIII-D over the last 16 years [39]; since 1980 confinement has doubled every two years. This improvement on performance is the result of two main factors: 1) the improvement on the device infrastructure (heating power, wall-conditioning techniques, divertor configuration, etc.) and 2) the skillful use of this technology by the experimentalists in controlling profiles and plasma conditions. As a result, enhanced confinement regimes have been accessed. In entering these new regimes, plasma transport properties change and plasma core confinement improves. These changes in confinement properties have provided us with new ways of understanding plasma transport. It is through such a process that the most significant progress has been made.

In this section, we will summarily review some of the most important changes on infrastructure that impact transport studies. The infrastructure is directly related to the four elements affecting transport as discussed in the introduction (sources, transport coefficients, boundary conditions, and flux geometry) and to its understanding (diagnostics).

A. Heating and Current Drive

Before the 1970's, plasma heating was part of the plasma production process. Auxiliary heated experiments—that is, experiments with auxiliary heating power larger than the

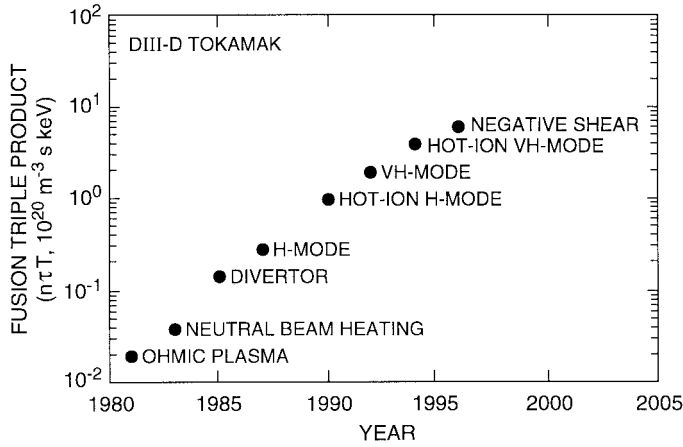


Fig. 2. Evolution of the best performance discharges in DIII-D over the last 16 years [38].

power used in creating the plasma—only started in the 1970's. Early in that decade, neutral beam-heating injection (NBI) was applied to heat tokamak plasmas [40], [41]. A beam of energetic neutral particles is injected, the neutral particles are ionized, and the fast ions transfer energy and momentum to the plasma through collisions. The heating mechanism is well understood [42] within the framework of classical theory. A major accomplishment for this heating technique was the high temperature reached in the PLT experiment. Plasmas with ion temperature up to $T_i = 20$ keV and with good confinement were obtained [43]. Most major tokamaks and stellarators have since incorporated NBI as a heating technique. Radio frequency (RF) is also used to transfer energy to the plasma. The use of RF heating in plasma experiments started before the 1970's, but no high-power (greater than ohmic) applications were carried out until the 1970's. Electron cyclotron (ECH), ion cyclotron, lower hybrid, and Alfvén wave heating are now used in many devices with success [44]. Ion Bernstein waves have also been effectively used [45]. The power increase and diversity of heating sources have been essential in accessing and understanding enhanced confinement regimes because some of these regimes have a power threshold and require profile control.

RF waves and NBI [46] are also used to drive plasma current in a tokamak. NBI current drive was first tested in the Levitron [47] and after that in several tokamaks. RF current drive experiments were carried out successfully in Alcator with lower hybrid current drive [48]. Lower hybrid current drive is effective in driving current near the plasma edge. Fast waves penetrated deeper into the plasmas; they are good candidates to drive current at the plasma core and have been proven effective in recent experiments [49], [50]. Off-axis current drive with ECH [51], [52] and mode-conversion current drive [53] have also been demonstrated. The high radial localization of the ECH power deposition makes its use very convenient for profile control. A goal of current drive is the sustainment of the tokamak current. Sustainment for about an hour has been demonstrated in the TRIAM-1M tokamak [54]. Up to 3.6 MA of the tokamak toroidal current has been driven with lower hybrid in JT-60U [55]. High-energy neutral beams can

also be used to drive current near the magnetic axis. Negative ion beams with energies of 500 keV have been developed and tested in JT-60U [8]. Combination of neutral beams and various RF systems is expected to allow full control and sustainment of the toroidal current in tokamaks. The progress in current drive will soon allow the current control needed to test ability of sustaining enhanced confinement regimes.

Active-feedback control systems for plasma position have greatly improved the tokamak operation over the early devices in which passive control was through a copper shell. This control is particularly important at high-power (high β) operation. Plasma position can be preprogrammed, allowing it to remain fixed or to be changed during a plasma discharge. This control has eliminated the random interactions of the plasma with material surfaces and the concomitant generation of impurities.

The combination of electron cyclotron heating for start up followed by neutral beam heating allowed the first stellarator finite-beta experiments with zero-current [56], [57]. These heating capabilities allowed the stellarator to become a confinement device competitive with tokamaks.

Apart from the obvious goal of heating the plasma, the use of auxiliary heating techniques and a good knowledge of the power deposition profiles provide us with powerful tools for transport studies. The determination of the power deposition for RF has taken more years to develop than for neutral-beam heating. Particularly for wavelengths comparable with machine size, full two-dimensional (2-D) calculations are required. It is now possible to use different sources of heating that allow the separation of electron and ion channels and also the heating effects from momentum-induced effects. Even more important, by combining heating and current drive techniques, it is now possible to control the temperature and current profiles, which is essential in accessing, understanding, and hopefully sustaining enhanced confinement regimes.

B. Plasma Fueling

Normal fueling of the plasma takes place at the edge through gas injection and recycling from the walls. In this situation, all particle sources are at the plasma edge; therefore, it is difficult to control the density profile. When NBI is used, there is central fueling by thermalization of the fast ions. Again, the fueling caused by the neutral beams is difficult to control. To have direct-controlled core fueling, pellet injectors [58] were developed during the 1970's and their technology improved over the following decades. By injecting high-speed pellets, it was possible to fuel the plasma core directly. In Ohmic-heated plasmas in Alcator, peaking the density profile led to improved confinement and the recovery of the positive scaling of confinement time with density [59]. The pellet injection technology has remained a powerful tool for density profile control in exploring new confinement regimes or in testing physics that depends on the details of the density profile. The use of high-field pellet injection is a hopeful development toward core fueling of reactor plasmas [60]. The use of compact torus injection as an alternative technology for core fueling is under development [61]. Initial testing of this fueling technology has been done in the Tokamak de Varennes [62].

C. Particle Exhaust and Wall Conditioning

A diverse array of techniques (poloidal divertors, stochastic divertors, pump limiters) for particle and energy exhaust have been explored. This is an area of intense research due to the need for development of technology for the next generation fusion devices [63] for controlling fueling and exhaust of helium ash. Apart from the obvious motivation for developing it (transport of particles and heat to the divertor chamber instead of to the vessel) [64], [65], the impact of particle control on the core plasma confinement has been very important by reducing the neutral population and impurities in the core plasma. Active pumping is also essential for particle control, particularly in high-power NBI heated discharges.

Numerous techniques for wall conditioning have been developed [66], from the early discharge cleaning techniques and gettering to boronization [67] and Lithium pellet injection [68]. It is universally accepted that wall conditioning is necessary for good confinement and for access-enhanced confinement regimes. However, we do not have a detailed quantitative knowledge of the connection between the conditioning techniques and plasma core transport. It is clear that control of the neutral population and impurities improves plasma conditions and results in better confinement; but we do not have a theoretical understanding nor a quantitative model of how the different wall-conditioning techniques produce the change on core plasma confinement. This remains a major issue in interpreting plasma transport experiments.

D. Diagnostic Development

Good diagnostics are essential in understanding plasma behavior and in particular plasma transport [69]. Many specialized diagnostics are needed for plasma transport studies. There are good review papers on plasma diagnostics [70], [71] where the reader can find a description of their physics basis. Here, we briefly review some of the diagnostics that have had a major impact on transport studies. Only ten years ago, it would have been impossible to predict the accuracy and sophistication of the present-day diagnostics. As a measure of the change in diagnostics, we can just compare the determination of plasma parameters in T-3, as described in the previous section, with the current set of diagnostics on a present-day device. To illustrate this point, I have listed in Table I the TFTR diagnostics.

There are three main types of measurements for transport studies: 1) determination of plasma equilibrium quantities; 2) characterization of plasma fluctuations; and 3) perturbative transport studies. Now, tokamak equilibria can be reconstructed quite accurately at different times during a discharge. This is due to the progress on the profile diagnostics. Some examples are the following.

- Electron temperature and density profiles with Thomson scattering. From the measurement of a single point per discharge in the early 1970's, we have progressed to multiple profiles for discharge. The accuracy of the detectors has also increased. Thomson scattering measurements are usually complemented with electron temperature measurement using electron cyclotron emission (ECE) and

density measurements with interferometry and reflectometry.

- Ion temperature profile diagnostics. Charge exchange recombination has allowed accurate determination of the ion temperature profile. This measurement has been critical in determining that the ion channel losses are well above neoclassical theory predictions and in proving that the anomalous ion transport is the dominant loss channel in L-mode discharges [72]. This diagnostic has also allowed the accurate time resolution of the electric field profile at the plasma edge, which is essential in identifying transition mechanisms to enhanced confinement regimes [73].
- Poloidal magnetic field measurement and determination of the safety q -profile and current density profile. It is accurately measured by using the motional Stark effect (MSE) [74]. This profile information is essential for a precise reconstruction of the tokamak equilibrium [75]. The MSE diagnostic plays a critical role in the study of discharges with a nonmonotonic safety factor q -profile, the reverse magnetic shear discharges. Now, the possibility of using this diagnostic to directly measure also the radial electric field is being investigated [76].

If the cause of anomalous transport is plasma turbulence, the understanding of the transport mechanism requires not only the detailed measurement of the profiles but also the measurement of the fluctuations and their relative phases. These measurements must be done over the relevant time and space scales. As a result, the difficulty of the measurement as well as the complexity of the data-handling systems and analysis are greatly increased. Early techniques for measuring plasma fluctuations are edge probes to measure plasma edge potential and density fluctuations, magnetic loops for poloidal magnetic field fluctuation, and infrared scattering for core density fluctuations. The development and application of new diagnostic tools for the fluctuations studies was stimulated by the Transport Initiative in 1988 [25]. Examples of fluctuation diagnostics techniques are the following.

- Heavy ion beam probe (HIBP) [77]. This diagnostic system injects a heavy ion beam through the plasma, the ions are further ionized and their outgoing orbit carries out information on the potential and local fluctuating quantities. HIBP has been successfully used in the determination of the radial electric field profile in tokamaks [78], [79], stellarators [80], and mirrors. It is also used to measure electrostatic potential and density fluctuations. Measurements of the plasma core fluctuation have been done in several devices [81]–[84], but the relative phase between the plasma potential and density fluctuations has only been measured in a few instances and close to the plasma edge.
- Beam emission spectroscopy (BES) [85]. Radiation intensity from excited beam atoms is proportional to local density and $\tilde{n}/n$ can be determined. The Doppler shift isolates radiation from the beam neutrals and a crossed beam provides localization. BES is used to measure the long wavelength end of the density fluctuation spectrum.

TABLE I
PLASMA DIAGNOSTICS ON TFTR

Diagnostic System	Configuration
Electron Temperature and Density TV Thomson Scattering (TVTS) ECE Heretodyne Radiometer ECE Fourier Transform Spectrometer ECE Grating Polychromator Multichannel Far Infra-Red Interferometer (MIRI)	Two midplane, 76 point radial points, horizontal midplane scan, 4 ms scan, 4 - 10 cm res. 15 ms scan, 2 - 6 cm res. 20 point horizontal midplane radial profile 10 vertical chords
Ion Temperature X-ray Crystal spectrometer Charge Exchange Recombination Spectroscopy (CHERS)	horizontal chord (point) 18 channel array
Impurity Concentration Visible Bremsstrahlung Array VuV Survey Spectrometer (SPRED) Multichannel Visible Spectrometer Pellet polychromator X-ray Pulse Height Analysis (PHA)	16 chord tangential horizontal array, 18 chord (HAIFA) poloidal array. 4 toroidal locations radial view in horizontal midplane 4 views with fiber optics 1 vertical location, 1 horizontal view horizontal chord
Radiated Power Tangential Bolometers Bolometer Arrays Wide-Angle Bolometers	16 chord counter-viewing, 2 chord co-viewing array 19 horizontal radial view, 19 vertical. chords 6 toroidal locations
Neutrons and Alpha-particles Epithermal Neutrons Neutron Activation Detector 14 MeV Neutron Detectors Neutron Fluctuation Detector Collimated Neutron Spectrometer Multichannel Neutron Collimator Lost Alpha/Triton Array Fast Neutron Scintillation Counters Gamma Spectrometer Alpha-Charge-Exchange Analyzer Alpha-Charge-Exchange Recombination Spectroscopy (α -CHERS) Gyrotron Scattering System Alpha-Charge-Exchange spectroscopy	6 toroidal locations with different sensitivities 7 toroidal locations, with 1 reentrant top location 1 toroidal location 6 locations 1 toroidal location 10 vertical channels; D-D and D-T sets of detectors 3 poloidal detectors, 1 movable near-midplane probe Horizontal midplane Horizontal view Horizontal midplane view (with Lithium Pellet Injector) 5 Horizontal radial channels, 5 vertical sightlines Small angle scattering, single mid-plane position, moveable between shots. Horizontal midplane view (with Lithium Pellet)

(a)

It has shown that the fluctuation spectrum peaks at long wavelength ($k_{\perp} \rho_s < 0.2$) and that the characteristic radial correlation of the density fluctuations at the tokamak core is of the order of 1–2.5 cm [86]. This technique has also been used to measure ion temperature fluctuations and velocity fluctuations [87].

- Reflectometry for measurement of density fluctuations [88]–[90]. Density variations give a phase modulation to small angle scattered radiation with respect to direct reflection with the main effect near the cutoff layer. Although the concept is not new, in the last few years it has been perfected to study plasma core density fluctuations and the determination of their radial correlation length [91], [92]. It has also been effective in showing fluctuation suppression by transport barriers [93].

- Plasma core electron temperature fluctuation measurements by correlation measurements of radiation emitted at electron cyclotron (ECE) harmonics [94], [95]. To measure the electron temperature fluctuations, it is necessary to extract the plasma fluctuation from the thermal noise. This requires careful correlation measurements. It has been found that long wavelength fluctuations are present at the plasma core with temperature fluctuation amplitudes comparable to the density fluctuations [96].
- Plasma edge electron temperature fluctuations using probes. Although probes have been used for a long time [33], [97], only recently they have been systematically applied in the study of temperature fluctuations. Several different techniques [98]–[100] have been tried, and comparisons between the different techniques are

TABLE I (Continued.)
PLASMA DIAGNOSTICS ON TFTR

Diagnostic System	Configuration
Magnetic Properties Rogowski Loops Voltage Loops B_θ, B_ϕ Loops Diamagnetic Loops Locked mode Coils	2 toroidal locations 6 poloidal locations and various saddle coil 2*26 pairs external to the vacuum vessel 2 toroidal locations 4 Toroidal locations
Plasma Edge/Wall Plasma TV IR Camera Filtered Diodes (H-alpha) Filtered Diodes (C-II) Sample Exposure Probe Disruption Monitor (IR Detector) Edge density reflectometer	periscopes at 2 locations periscopes at 2 locations 5 telescopes in poloidal array 5 telescopes in poloidal array 1 edge location, moveable radially between shots 1 toroidal location 1 channel which is scannable to get edge n_e profile
Fluctuations/Wave Activities Microwave Scattering X-mode Microwave Reflectometer Beam Emission Spectroscopy X-ray Imaging System ECE Grating Polychromator Mirnov Coils Neutron Fluctuation Detector RF Probes	Scannable antenna and receiver 4 variable frequency system at 1 toroidal location 25 channel variable radial and poloidal profile at 1 toroidal location 20 vertical chords, 64 horizontal chords at 1 toroidal location 2 toroidal locations with 20 point radial profile 40 locations inside vacuum vessel 6 toroidal locations 7 toroidal locations
Plasma Current/Safety Factor Profiles Motional Stark Effect Polarimeter	10-channel radial profile at 1 toroidal location
Miscellaneous Hard X-ray Monitors Torus Pressure Gauges Residual Gas Analyzers Vacuum Vessel Illumination	5 wall locations 2 toroidal locations 2 toroidal locations on pump ducts 3 top toroidal locations

(b)

underway. Experimental results indicate that edge temperature fluctuations can be comparable to density fluctuations, but are seldom larger. The measurement of the temperature fluctuations is essential for the accurate determination of the turbulent fluxes at the plasma edge [101].

- Probes to measure the magnetic-induced components of the particle and energy fluxes. The latter consist of a fast pyrobolometer and magnetic probe to correlate the fluctuating parallel heat flux with $\tilde{B}_r$. These diagnostics have been developed by the MST group [102] and applied to RFP, stellarators, and tokamak edge plasmas.
- Phase contrast imaging diagnostic [103] is being used for 2-D imaging of density fluctuations. This diagnostic could extend the range of density fluctuation measurements at the longer wavelength range of the k -spectrum.

A fundamental issue in transport is the local character of transport or, equivalently, how well represented plasma transport is by equations of the form of (1)–(4). Perturbative experiments play a very important role in testing these

assumptions and in finding the form of the anomalous fluxes and the parameter dependences of the transport coefficients. The first perturbative experiments were carried out in the late 1970's [104], but in recent years we have seen more systematic perturbative studies [105]. To carry out these experiments, we need diagnostics that measure density and temperature in a continuous way at different radial locations and active diagnostic systems to produce localized perturbations. Examples of these active diagnostics are impurity injection by laser blow off, ECH heat pulse or a naturally generated plasma perturbation (sawtooth drop, L to H transition, etc.). The ECE measurements solve this problem for electron temperature (soft X rays have been also used but are more difficult to quantify). These types of experiments have raised serious questions on the local nature of the anomalous transport [106], as we discuss in Section VI.

Although progress in diagnostic development has been very significant, a complete set of diagnostics that covers the needs for transport understanding does not yet exist. The most obvious needs are the determination of the relative phase between

core fluctuations (HIBP is in principle capable of doing so, but there is none installed in any of the major tokamaks) and core plasma magnetic fluctuations. The latter have been measured in Tore Supra using a cross-polarizing scattering diagnostic [107] and larger scale magnetic fluctuations with HIBP in TEXT [108]. There are also indirect methods of measuring the core magnetically induced transport, such as the use of runaway electrons [109], [110]. The interpretation of the results from experiments based on runaway electrons is not simple, and requires a refinement of the theoretical models for particle diffusion in stochastic magnetic fields. These diagnostics should be utilized in other devices to systematically study the potential role of magnetic fluctuation on transport.

E. Development in Computer Capabilities

In the last 25 years, the most dramatic change in infrastructure is possibly in the area of computers. From the relatively small memory and low-speed sequential computers of the early 1970's, we have progressed to high-speed and large-memory computers of the 1990's, evolving through vector-, parallel-, and massive parallel-processing. This evolution has totally transformed the calculation capabilities and opened the possibility of relevant turbulence calculations—the relevance being not only the needed high resolution for separation of disparate time and space scales, but also the incorporation of realistic geometry, which is essential for the simulation of toroidal confinement experiments. In spite of the great progress made, a full numerical treatment of the plasma turbulence in toroidal geometry will require a future generation of computers. As a result of the evolution on computer capabilities, computational science has been established as the third branch for scientific inquiry, together with theory and experiment. Numerical studies of turbulent transport tokamak are the core of the Numerical Tokamak Project [111], [112], one of the DOE's Grand Challenge Problems.

Detailed computation of 3-D magnetic configurations has been essential in the design of modern stellarators. It has played an essential role in accurate coil design and in eliminating magnetic field errors that were a major cause for poor performance in early designs. It also has been critical in the optimization of the magnetic field components that has led to a substantial reduction in the neoclassical transport in those configurations [113].

The computer development has not just extended the possibilities of carrying out complex nonlinear calculations, but also has considerably increased the data handling and analysis capabilities. It is now possible to handle the large amounts of data that results from the sophisticated diagnostic systems and process it at relative high speed. This allows partial analysis of a discharge in between shots and provides important feedback to guide the experiment.

IV. PROGRESS IN THE EXPERIMENTAL TRANSPORT STUDIES

Experimental progress in confinement has been extensive and is difficult to summarize in a few pages. In this section, I have chosen a few areas of experimental research that illustrate

the increase in confinement performance and of empirical understanding of the transport phenomena.

A. Enhanced Confinement Regimes

Although there have been some ups and downs in the road, the increase in confinement performance during the last 25 years has been spectacular. The gloomy predictions for confinement reductions at high temperatures that were made in the early 1970's on the basis of trapped-particle modes were not verified. The results from PLT, with $T_i = 20$ keV [43], indicated the absence of such deleterious confinement effects in high-temperature plasmas. A few years later, this optimistic result was followed by the realization in ISX-B [114] that confinement deteriorated with increasing power. This was clearly reflected in the ISX-B scaling law $\tau_E \propto I_p^{3/2} P_b^{-2/3}$. Here, I_p is the plasma current and P_b is the neutral beam power. This confinement mode was confirmed in most experimental devices and it was later characterized by Goldston scaling [115] of the energy confinement time

$$\tau_E = 0.03 I_p R_0^{1.75} a^{-0.37} \frac{\sqrt{\kappa}}{\sqrt{P_{\text{tot}}}}. \quad (9)$$

Here, τ_E is measured in seconds, I_p in MA, P_{tot} in MW, R_0 and a in m , and κ is the plasma cross-section elongation. This mode of tokamak operation has been called L-mode. Recent systematic analysis of L-mode experimental data has produced more accurate scaling laws, but they do not change qualitatively the scaling that shows a nearly linear positive dependence on current, a decrease of confinement with increasing power, and very weak or no dependence on density and magnetic field. The scaling law used as a reference for L-mode confinement is the so-called ITER89-P [116]

$$\tau_E = 0.048 M_i^{0.5} I_p^{0.85} R_0^{1.2} a^{0.3} \kappa^{0.5} \bar{n}^{0.1} B^{0.2} P_{\text{tot}}^{-0.5}. \quad (10)$$

Here, M_i is the isotopic mass number and $\bar{n}$ is the line-averaged density in units of 10^{20} m^{-3} .

From the beginning of the 1980's, confinement experiments have pursued a path of continuous improvement in confinement. The impact of hardware improvements and wall-conditioning techniques [117] as listed in the previous section, together with the increased knowledge on how to control discharges profiles, has enabled access to new enhanced confinement regimes. A partial list of these improved confinement modes with some selected references are provided in Table II. In this table, the given transport enhancement factor H is the ratio of the confinement time in the enhanced confinement mode to the L-mode confinement time (as given by the ITER89-P scaling law). This factor is only a rough measure of the improvement for the best discharges. In each confinement mode, H should be a function of the relevant dimensionless parameters. An empirical classification of archetype confinement modes based on the main control parameter to access the enhanced confinement regime can be done, as indicated in Table III. This classification is an oversimplification because more than one control parameter is used to access most of the enhanced confinement regimes, and in each device, the conditions to achieve a confinement regime may be somewhat

TABLE II
IMPROVED CONFINEMENT REGIMES IN TOKAMAK PLASMAS

Confinement Mode Description	Change on Transport	Effect on Fluctuations
High Confinement Mode (H-mode) [118-120].	Edge transport barrier formation. Reduction of χ_i , χ_e , and D $H \approx 2$.	Change of $\bar{n} - \bar{\phi}$ phase followed by fluctuation suppression at the plasma edge [321]. Reduced radial correlation at the core [92].
Impurity Modes: Z-mode [160], I-mode [161], and RI-mode [162].	Reduction of D and χ_e in neutral beam heated plasmas $H \approx 1.8$.	Reduction of $\bar{B}_0/B_0$ no change in $\bar{n}/n$.
Pellet Enhanced Performance (PEP mode) [59] use pellets to peak density profile may be combined with H-mode [455].	Recovers Alcator scaling χ_e decrease by 3, χ_i reduced to neoclassical [456].	?
Improved Ohmic Confinement (IOC) [457] Peaked density.	Recovers Alcator scaling.	?
Hot Ion Modes: Supershot [458], I-mode [161], and other. Sometimes combined with H-mode [459].	Reduction of χ_e the most, also χ_i , and χ_ϕ . $\chi_e > \chi_i$ $H \approx 2.0$.	Slow increase of $\bar{n}/n$ and reduction of its radial correlation length with power [91].
Very High Confinement Mode (VH-mode) [144, 145]: extension of the H-mode to the plasma interior.	Broadens the edge transport barrier formation. Reduction of χ_i , χ_e , and D $H \approx 4$.	Disappearance of density fluctuation bursts [146].
Internal confinement barriers [147] and Core H-mode [148] (induced by Ion Bernstein wave heating).	Reduction of χ_i , and χ_ϕ . No change in χ_e .	?
High β_p modes [155], sometimes combined with H-mode [156].	High β_p mode: $H \approx 2-3$ High β_p H-mode: $H \approx 3.6$.	Reduction of $\bar{B}_0/B_0$ [460].
High l_i modes [152-154].	$H \approx 3$. $H \propto l_i$.	?
Reversed Shear mode (RS) [157-159].	$H \approx 2.5$.	Reduction of $\bar{n}/n$ with quasiperiodic bursts [149].
Enhanced Reversed Shear (ERS) [5] and Negative Central Shear (NCS) [6],[7] Modes, ITBNS [8].	Internal transport barrier formation. Neoclassical χ_i , and D . Reduction of χ_e .	Suppression of fluctuation at the core [149].

different. Many of the enhanced confinement regimes are a combination of archetype modes from Table III. The physical mechanisms thought to be responsible for the confinement improvement are discussed in Section V.

The first and probably the best studied of the enhanced confinement regimes is the H-mode [118]–[120]. It was first identified in the ASDEX tokamak [121] and then reproduced in most confinement devices. One of the more important properties of the H-mode is its universality. It has been accessed in different confinement devices (tokamaks, stellarators, mirrors), with different types of heating sources (ohmic, NBI, RF), with different boundary geometry (separatrix, limiter, circular, and shaped plasmas), and it can be spontaneous or externally triggered [122]–[126]. The universality of this confinement

mode points to the existence of a physical mechanism that is responsible for the H-mode confinement. This physical mechanism is discussed in the next section.

There is a clear transition from the L-mode to the H-mode with the formation of a plasma-edge transport barrier. In the region of the transport barrier, there is a reduction on the fluctuation level and a build up of the density and temperature gradients (Fig. 3). This transition has been characterized by a power threshold. From the first experiments, the power threshold P_{th} has been characterized by a positive power dependence on density and magnetic field [127]–[129]. The first empirical scaling law was of the form $P_{th} \propto \bar{n}B$. This scaling is found for single-null divertor configurations with the ∇B -direction toward the X -point. Changes on the divertor

TABLE III
CLASSIFICATION OF ARCHETYPE ENHANCED CONFINEMENT REGIMES IN TERMS OF THE CONTROL PARAMETERS

Wall Conditioning	Edge control	H - mode	E_r - profile control? →	{ ERS - mode NCS - mode
	n - profile control	Z - mode		
	T_i - profile control	PEP - mode		
	V - profile control	IOC		
	J_ϕ - profile control	Hot - ion mode		
		Supershot		
		VH - mode		
		Internal barrier		
		CH - mode		
		High I_i - mode		
		High β_p - mode		
		RS - mode		

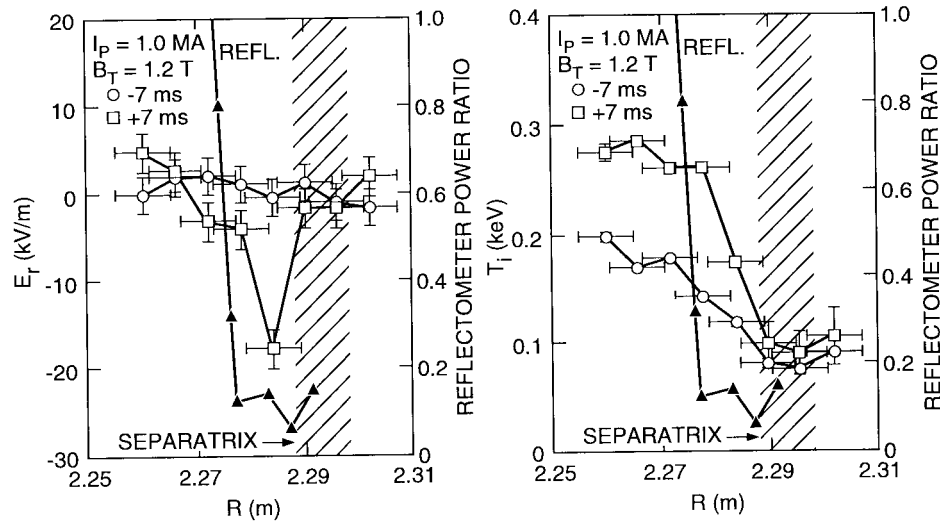


Fig. 3. Reduction on the fluctuation level and a build up of temperature gradients in the region of the transport barrier at the L-mode to the H-mode transition [71]. The E_r and T_i profiles are shown 7 ms before and 7 ms after the transition near the separatrix. The ratio of the fluctuation level after the transition to the level before is also plotted.

configuration and ∇B -direction can lead to totally different scaling laws [130]. Now, there is a relative abundance of data from many devices, in particular a database for ITER has been created. These data permit the refinement of this scaling law. Variants have been obtained by requiring dimensionally correct parametric dependences [131]

$$P_{th} = 0.45 B n^{0.75} R^2. \quad (11)$$

It is not clear that expressing the power threshold in term of plasma physics dimensionless parameters is the correct approach. Since the L-H transition is a plasma-edge phenomena, atomic physics can play an important role in its dynamics [132]–[134]. Therefore, some parameter reflecting the atomic physics may be needed in determining the parametric dependence of the power threshold. For similar reasons, power is probably not the right parameter to characterize the transition, local-edge parameters can provide a better characterization of the threshold. Considerable effort is now dedicated to ex-

perimentally identify these local-edge parameters [60], [135], [136].

The time of the transition is empirically identified with a drop in the D_α (H_α) signal. The transition time varies and ranges from very fast (a few hundreds of microseconds), to very slow (a few tens of milliseconds). Dithering transitions are also observed. In these transitions, the edge plasmas oscillate between L and H modes before making the final transition to H-mode [137], [138].

The edge transition is followed by confinement improvement at the plasma core. There is no experimental agreement on how fast the confinement improvement propagates into the core. In DIII-D, the propagation seems compatible with a diffusive process, while in JET it is characterized by being much shorter than the profile evolution time [139]. The overall confinement enhancement factor H varies with plasma conditions, but its averaged value is about two. There is clear evidence that H-mode is a different confinement regime than the L-mode. The H-mode confinement is not just a quantitative

increase in core confinement, as there is a qualitative change in confinement properties. For instance

- empirical confinement laws show different dependence of the global confinement time with the plasma parameters. We can compare the basic L-mode scaling (10) with the empirical scaling for ELM-free H-mode confinement [140]

$$\tau_E = 0.053 I_p^{1.06} B^{0.32} \bar{n}^{0.17} M_i^{0.41} R^{1.68} a^{-0.11} \kappa^{0.66} P^{-0.67}. \quad (12)$$

Here, we have followed the same normalization conventions as (9):

- the local-heat diffusivity shows a gyroBohm scaling instead of Bohm [141];
- radial correlation lengths of the turbulence are shorter in the H-mode (≈ 0.5 cm) than in the L-mode (≈ 1 – 2 cm) [92].

After profiles have relaxed in the H-mode, quasiperiodic relaxation oscillations are often observed at the plasma edge. They are called edge-localized modes (ELM) [142], [143]. They may play the positive role of periodically flushing impurities out of the core, but this large burst of particles in the divertor may cause damage to the divertor plates.

The transition from H-mode back to L-mode is characterized by a lower power threshold than the direct transition. The power threshold for the back transition is poorly known. However, the fact that the back transition has a lower power threshold indicates hysteresis behavior. The experimental characterization of threshold should be done in terms of local-plasma-edge parameters, in order to compare with theoretical models for the transition. Many of the enumerated properties of the L–H transition are similar to a phase transition in a solid.

The very high confinement mode (VH-mode) [144], [145] is a natural continuation of the H-mode. To access the VH-mode regime requires a quiescent H-mode (no ELM activity). The transport barrier is able to penetrate deeper into the plasma with a concomitant increase in the confinement time. The increase of confinement time and barrier penetration coincides with an increase in the toroidal velocity gradient [146]. It is not clear if different devices apply to the same criteria in differentiating between H-mode and VH-mode.

There are several other confinement modes that are characterized by the formation of internal transport barriers [147]. The possibility of creating transport barriers in the plasma core opened a new path to improving confinement. The first internal barrier experiments in JT-60 indicate a link between the barrier position and low- q rational surface ($q = 3$). These experiments showed that transport barriers are not an intrinsic property of plasma edge. It also suggested the possibility of controlling transport barriers. Of particular interest is the core H-mode (CH-mode) [148] which was induced by ion Bernstein wave heating, thus offering a way of externally controlling transport barriers.

In recent experiments, a high-confinement regime with internal transport barrier has been accessed by carefully controlling the current (creating a region with negative magnetic shear at the plasma core) and tailoring the pressure profile (to increase

gradients without causing a disruption) [7]. This confinement mode with strong rotation shear is called the negative central-shear mode (NCS) [6], [7] in DIII-D, the enhanced reversed-shear mode (ERS) [5] in TFTR, and the internal-transport barrier with negative magnetic shear (ITBNMS) in JT-60U [8]. In TFTR and with this mode of operation, anomalous transport at the plasma core is practically eliminated in both the ion thermal and particle channels. The anomalous electron thermal diffusivity is reduced but not always suppressed. There is a clear transition to this mode of operation, indicating the existence of bifurcated states as in the H-mode case. The fluctuation diagnostics indicate a very quiescent plasma during these enhanced confinement modes [149]. In DIII-D and JT-60U, effective transport barriers in the electron channel have also been created. This suggests the existence of different transport mechanisms in electron and ion. Electron transport barriers have also been created in Tore Supra [150] and W7-AS [151] under different heating scenarios.

Other confinement modes are not necessarily accessed through a sharp transition. For instance, access to high I_i [152]–[154] and high β_p modes [155], [156] depend on how the current profile is tailored. In this family of operation modes, we should include the reverse shear (RS) [157]–[159] mode which can be the prelude to the ERS/NCS confinement mode.

Identifying the physics mechanisms in operation in these confinement modes is also complicated because some of them combine features of several archetype confinement modes. For instance, the JET PEP modes are combined with an edge H-mode and have reverse magnetic shear at the center of the plasma [159]. Some of the highest performance discharges in JT-60U have been obtained by combining high β_p mode with H-mode.

An alternative approach to the tokamak confinement has been open through a sequence of improved confinement modes coupled with edge radiation. The increased edge-plasma radiation does not reduce the confinement, but it can improve it. This was already shown in the case of the Z-mode [160]. In TEXTOR, by combining a radiative edge with an improved confinement mode (I-mode) [161], an interesting new regime, the radiative I-mode (RI-mode) [162], has been accessed. This regime combines high confinement, high β , low-edge q values and high density with dominant-edge radiative-heat exhaust. This mode of operation could eliminate the dilemma of having a hot plasma edge and wanting to minimize the energy fluxes to the divertor plate.

The fluctuation characterization of some enhanced confinement regimes is rather incomplete, as can be seen from Table II. Understanding transport requires a systematic study of the fluctuations in those regimes.

B. Characterization of the Confinement: Scaling Laws and Dimensionally Similar Discharges

Since the first confinement experiments, empirical information on transport properties has normally been translated into scaling laws. In the 1970's and early 1980's, these scaling laws were derived for the energy confinement time. It was assumed

that the energy confinement time is a function of the form

$$\tau_E \propto B^{\alpha_1} n^{\alpha_2} R^{\alpha_3} a^{\alpha_4} \kappa^{\alpha_5} P^{\alpha_6} q^{\alpha_7} \dots \quad (13)$$

in terms of the experimentally controllable parameters, and the exponents were determined from the data available from a single machine or from many experiments.

To empirically characterize the local diffusion coefficients, it is important to consider constraints resulting from general invariance principles [163]–[165]. These constraints reduce the number of experiments needed to determine the coefficients. To establish these constraints, we need to identify the relevant dimensionless parameters and the invariance principle. This can only be done within some given theoretical framework. The broader the framework, the less constrained the transport coefficients. On strict theoretical grounds, the number of dimensionless parameters for a confined plasma is large, up to 19 have been identified [166]. They include plasma physics parameters such as β , the ratio of the plasma kinetic pressure to the magnetic pressure, the collisionality, $\nu_* \equiv \nu_{ei}/\omega_b$, and the normalized Larmor radius, $\rho_* \equiv \rho/a$. Here, ν_{ei} is the electron-ion collision frequency, $\omega_b = V_{||}/(Rq)$ is the bounce frequency, $\rho = V_{\perp}/\Omega$ is the Larmor radius, and $\Omega = eB/mc$ is the cyclotron frequency. There are also parameters describing the magnetic field geometry, like q , aspect ratio $A \equiv R/a$, and the ellipticity κ and triangularity δ of the plasma cross section and parameters representing the plasma composition T_e/T_i , m_e/m_i , Z_{eff} , $\dots$. For a local diffusion coefficient, we have to include the parameters related to plasma profiles, like the ratios of scale lengths L_T/R , L_n/R , $\dots$. From dimensional arguments, the diffusivities can be expressed in the form

$$\chi = c_s \rho_*^\alpha F(\nu_*, \beta, q, A, \kappa, \delta, L_T/R, L_n/R, \dots, m_e/m_i, T_e/T_i, \dots). \quad (14)$$

The corresponding form for the global confinement time is

$$\tau_E = \frac{1}{\Omega} \rho_*^{-(2+\alpha)} G(\nu_*, \beta, q, A, \kappa, \delta, L_T/R, L_n/R, \dots, m_e/m_i, T_e/T_i, \dots). \quad (15)$$

In these expressions, F and G are functions of the dimensionless parameters to be determined. Two limits that are often discussed in transport studies are defined by (14). When $\alpha = 1$, the scaling law is called gyroBohm. This is the expected scaling from most local turbulence theories for which the turbulent scale length Δ_T is proportional to ρ . For $\alpha = 0$, the scaling is called Bohm because χ scales like D_{Bohm} , although its value can be lower. These dimensional constraints also imply constraints on the global confinement time scaling. They lead to relations between the α_i exponents in (12).

In a given experimental device, the main control parameters are the magnetic field B (and/or I_p), the plasma density n , and the plasma temperature T (through the applied power). These parameters are simply related to three dimensionless parameters ρ_* , β , and ν_*

$$B \approx \left(\frac{q\beta M_i^3}{\nu_* \rho_*^6 a^5} \right)^{1/4} \quad n \approx \frac{\beta M_i}{\rho_*^2 a^2} \quad T \approx \left(\frac{q\beta M_i}{\nu_* \rho_*^2 a} \right)^{1/2}. \quad (16)$$

Using these relations, one can derive the scaling of the confinement time on the dimensionless parameters from the

empirical scaling laws. Hence, in the L-mode and base on the ITER89-P scaling [see (10)]

$$\Omega \tau_E \propto \rho_*^{-2.05} \beta^{-0.52} \nu_*^{-0.27} \quad (17)$$

and for ELM-free H-mode (12)

$$\Omega \tau_E \propto \rho_*^{-2.71} \beta^{-1.22} \nu_*^{-0.28}. \quad (18)$$

By expressing the confinement time in this form, it becomes apparent which are the parameters playing a critical role in confinement.

The first scaling studies in tokamaks were done for ohmic plasmas. One of the results that was reproduced in most ohmic-heated tokamaks is the linear increase of the energy-confinement time with density, the Alcator scaling [167]. This scaling law holds at low density in ohmic discharges. However, high-density experiments show that a saturation effect τ_E becomes independent of density. Several mechanisms have been proposed to explain the saturation effect, one of the most plausible being the onset of the η_i turbulence; this is discussed in the next section. The recovery of the $\tau_E \propto n$ scaling when the density profiles are peaked by pellet injection [59] supports this assumption. Equations (10) and (12) are examples of scaling laws for L-mode and H-mode, respectively. Scaling laws of the form of (12) are useful in comparisons between experiments and in projecting to new ones (with the caution that should always be used in extrapolating from empirical laws). In the middle of the 1980's, when JET, TFTR, and JT-60 started operation, the confinement time in L-mode operation was in very good agreement with the extrapolation from (9), which was derived with data from the previous generation of devices (Alcator, Pulsator, T-11, FT, ST, TFR, ISX-B, Doublet III, PLT, PDX, and ASDEX). This result gave credibility to the L-mode scaling law. Since no device has been built after those three, the predictive capabilities of other scaling relations for the improved confinement regimes have not yet been put to a similar test.

Although there may be a use for global confinement times in planning new experiments, they have little physics content. First, they combine plasma core with edge confinement, each with different underlying physics mechanisms. Second, we expect the local physics parameters to be the ones relevant for the local transport. A new experimental approach was proposed in determining empirical transport scaling laws of the form of (14) [168]. The experimental approach requires a series of experiments to have been performed to resolve the scaling of D with different dimensionless parameters, for instance with ρ_* . In a given device, tokamak discharges that have the same dimensionless parameters, but with different values of ρ_* , are produced. Naturally, these discharges have different engineering parameters. The comparison of the confinement times of these discharges gives the scaling with ρ_* . The degree of profile control required for these experiments makes them difficult. In a given device, only a limited range of variation on ρ_* can be achieved (usually about a factor of two is the maximum variation achieved). A larger range of variation in ρ_* can be obtained by comparing dimensionally similar discharges from different devices, but in this case the

experiments are even trickier. In spite of its difficulty, this approach has been proven to be quite effective.

At present and for L-mode, most tokamak experimental results are consistent with Bohm-type scaling [141], [166], [169], [170], while others are consistent with ρ_* scaling between Bohm and gyroBohm [171]. The empirical scaling of the global confinement time is also close to Bohm scaling, as shown in (12). However, the problem determining the scaling of the diffusivities is more complicated than stated because electron diffusivity may be different from ion diffusivity. The separation between both transport channels requires a diagnosis of the plasma and makes the analysis of the experimental results more difficult. Only a few experiments have been done on separating electron and ion channels. Experiments in DIII-D [141] separating the electron and ion channels indicate that Bohm scaling is the result of $\chi_e \approx \chi_{\text{Bohm}} \rho_*^{1.1 \pm 0.3}$, close to gyroBohm scaling, and $\chi_{\text{ion}} \approx \chi_{\text{Bohm}} \rho_*^{-0.5 \pm 0.3}$, worse than Bohm scaling and close to the ρ_* scaling given by Goldston scaling [see (9)]. The resulting scaling observed for the single-fluid thermal-transport coefficient will therefore depend on the relative dominance of the ion and electron channels.

The situation is more complicated for H-mode discharges. The scaling of the ion thermal diffusivity seems to depend on the proximity to the L to H-mode threshold. It remains close to Bohm scaling near the transition threshold and changes to gyroBohm scaling away from it. The electron thermal diffusivity maintains a basic gyroBohm scaling. Again, the global confinement time (12) also shows scaling close to gyroBohm scaling.

The ρ_* scaling is not the only one that has been experimentally investigated. Experiments testing the β and ν_* scaling have also been carried out. In L-mode, the thermal diffusivities show practically no dependence on these two parameters. For the ELMy H-mode, both the β and ν_* dependences are very weak [170], [172]. These results are in clear contradiction to the β -scaling of global confinement time in both regimes [see (17) and (18)]. Using these experimentally determined scaling exponents of the dimensionless parameters from the local diffusivities, it is possible to derive the scaling of the global confinement time in terms of the machine parameters. Relatively small changes in the power dependence of the confinement time may resolve the discrepancy. For instance, using the DIII-D results, the ELM-free H-mode scaling turns out to be $\tau_E \propto I^{1.0} B^{0.09} n^{0.27} P^{-0.45}$. Although the change of exponent is small, its impact on the design of future devices can be large. This discrepancy between the single dimensionless parameter scaling experiment and the one obtained from fitting a broad database can be due to ill conditioning of the data. The clarification of this issue is rather important.

The empirical determination of the scaling of the diffusivities with the relevant dimensionless parameters is a powerful tool not only in extrapolating to the next generation devices, but also in unraveling the nature of the transport mechanisms. For instance, the strong β degradation of the confinement suggested by the global scaling laws disagrees with the interpretation of the anomalous transport being dominated by electrostatic turbulence. If this dependence is not real, as local scaling seems to indicate, this removes one of the

issues in the interpretation of the existing data. Therefore, the determination of the scaling of the diffusivities goes beyond the characterization of the confinement, and it can give an indication of the underlying physics mechanism.

Finally, to test the basic principle of this approach, it is necessary to show that dimensionally equivalent discharges in different devices have the same transport properties. Experiments in several devices are underway and the initial results give support to this approach [172].

C. Characterization of the Plasma Transport by Fluctuations

A deeper level in the characterization of transport in magnetically confined plasmas is to ascertain whether turbulence is the underlying cause for anomalous transport [20], [173]. The first measurements of tokamak core fluctuations were scattering measurements in the ATC tokamak [174], [175]. They showed the turbulent nature of the confined plasmas. Since then, microwave scattering measurements have been used in many tokamaks and stellarators. Transport studies have required the development of other fluctuation diagnostics, as discussed in the previous section. All the fluctuation diagnostics have shown the turbulent character of confined plasmas.

In tokamaks and stellarators, the relative potential and density fluctuation levels at the plasma core are very similar, $\langle |e|\tilde{\Phi}/T_e \rangle^{1/2} \approx \langle (\tilde{n}/n)^2 \rangle^{1/2}$ and about 1% or less. The fluctuation levels increase toward the plasma edge where fluctuation levels can reach values of 20–60% [176]–[178]. In Fig. 4, we show an example of radial profile of fluctuations from ATF. The results from several diagnostics have been combined in plotting this chart [179]. The plasma edge in ohmic-type discharges is limited by a velocity-shear layer [180]. This velocity-shear layer determines the characteristic edge-confinement radius and provides a useful way of comparing results from different magnetic confinement devices. At the plasma edge, inside the shear layer, $\langle |e|\tilde{\Phi}/T_e \rangle^{1/2} > \langle (\tilde{n}/n)^2 \rangle^{1/2} \geq \langle (\tilde{T}_e/T_e)^2 \rangle^{1/2}$. However, outside the velocity-shear layer, the electrostatic potential fluctuations decrease while both $\tilde{n}/n$ and $\tilde{T}_e/T_e$ increase, as is shown in Fig. 5 from detailed edge measurements in the TEXT tokamak [181]. Although the underlying physics mechanisms for turbulence in ATF and TEXT may be quite different, the profiles of the fluctuation levels have strong similarity.

In going from ohmic to L-mode, core density and electrostatic potential fluctuation levels increase [83]. The character of turbulence seems to change. This change is manifested by an increase in the radial correlation length, which is about 0.4 cm in ohmic and monotonically increases with power in L-mode [92]. In the H-mode regime, the radial correlation length is like in ohmic plasmas. In L-mode, microwave scattering [182] and BES [87] measurements at the plasma core show that the $k_{\perp}$ spectrum increases monotonically toward the low- $k_{\perp}$ values long wavelengths.

The magnetic fluctuations also decrease from the plasma edge to the core, but we lack systematic data. For reverse field pinch (RFP) configurations, the electrostatic fluctuations have a similar radial dependence to tokamaks and stellarators,

TABLE IV
FLUCTUATION-INDUCED FLUXES

Turbulent fluxes	Electrostatic contribution	Magnetic contribution
Particle flux	$\langle \tilde{n} \tilde{V}_r \rangle$	$-\frac{\langle \tilde{J}_\parallel \tilde{B}_r \rangle}{e B_\phi}$
Energy flux	$n \langle \tilde{V}_r \tilde{T} \rangle + T \langle \tilde{V}_r \tilde{n} \rangle$	$\frac{\langle \tilde{Q}_\parallel \tilde{B}_r \rangle}{B_\phi}$

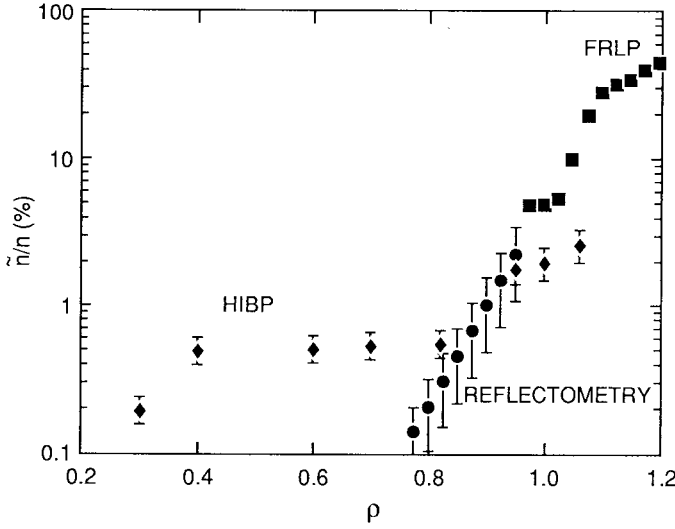


Fig. 4. Radial profile of density fluctuations (in %) in ATF stellarator obtained by combining results from different diagnostics [177].

however, magnetic fluctuations remain dominant at the plasma core.

Measurements at the plasma edge in the Continuous Current Tokamak (CCT) with a 2-D array of probes show the existence of complex large-flow structures [183]. It is not clear whether these structures exist in all tokamaks or occur in only a few devices. In the scrape-off layer (SOL) of ASDEX and W7-AS [184], fluctuations present strong poloidal asymmetry; asymmetries have also been measured at the plasma edge in T-10 [185]. These observations underline the importance of 2-D fluctuation measurements for the study of edge-plasma turbulence.

Even with all existing measurements, it is difficult to definitively answer the question: Is the anomalous plasma transport caused by plasma turbulence? One way of answering is by calculating the different fluxes induced by turbulence and comparing them to the particle and energy fluxes determined from the power balance. The fluctuation-induced fluxes are listed in Table IV, where we have only considered quadratic contributions in the fluctuating quantities. In this table, tilde indicates fluctuating quantities, the square brackets $\langle \rangle$ indicate ensemble averaged, $J_\parallel$ is the parallel current, $Q_\parallel$ is the parallel energy flux, and V_r is the radial component of the $E \times B$ velocity.

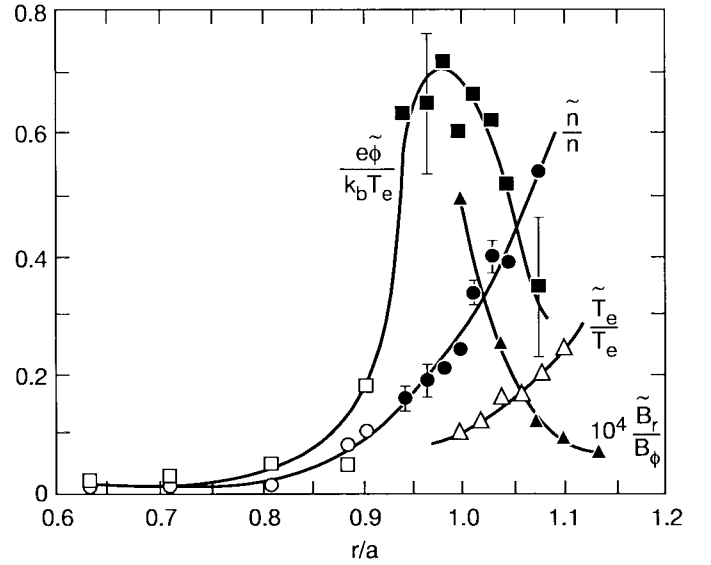


Fig. 5. Radial profile of fluctuation near the edge in the TEXT tokamak [178].

Most of the existing fluctuation diagnostics are directed toward electrostatic turbulence. Even if we only consider electrostatic fluctuations, to measure the transport induced by fluctuations requires not only measurement of fluctuation levels but also their cross-correlation or relative phase. For instance, in a tokamak, the electrostatic contribution to the particle flux is $\Gamma_p = \langle \tilde{E}_\theta \tilde{n} \rangle / B_\phi \equiv \langle \tilde{E}_\theta^2 \rangle^{1/2} \langle \tilde{n}^2 \rangle^{1/2} \cos \varphi_{En} / B_\phi$, where φ_{En} is the relative phase between $\tilde{E}_\theta$ and $\tilde{n}$. To calculate Γ_p , we need the fluctuating poloidal electric field and the density fluctuations. The fluctuating poloidal electric field can be determined from the measurement of the fluctuating plasma potential at two poloidal positions, $\tilde{E}_\theta = (1/r)(\partial \tilde{\Phi} / \partial \theta)$. The density fluctuations can be directly measured. The correlation between these two quantities determines the electrostatic contribution to the fluctuation-induced particle flux. Let us first consider the contributions to the particle flux

- At the plasma edge, probes can be used for the determination of the particle flux. Probes measure the floating potential and the saturation current, to extract the $\tilde{\Phi}$ and $\tilde{n}$, it is necessary to also know the electron temperature

TABLE V
IMPROVED CONFINEMENT REGIMES: CONJECTURED TRANSITION MECHANISMS

Confinement Mode	Possible Origin of E_r Barrier	Alternative and/or Complementary Mechanisms
High Confinement Mode (H-mode)	V_θ and ∇P_i Orbit loss	ω^*_i stabilization Magnetic shear stabilization
Impurity Mode: Z-mode, RI-mode	Edge perturbation	ω^*_e stabilization
Pellet Enhanced Performance (PEP mode)	∇n	η_i below threshold
Improved Ohmic Confinement (IOC)	∇n	η_i below threshold
Hot Ion Modes: Supershot, I-mode, etc.	∇P_i	Profile effects
Very High Confinement Mode (VH-mode)	V_ϕ and ∇P_i	$V_{ }$ Second stability
Internal confinement barriers and Core H-mode	IBW induced flow $\nabla P_i, V_\phi, V_\theta$	?
High β_p modes,	B_θ	Second stability Drift reversal
High I_i modes	$B_\theta?$	Shear stabilization at the edge and increased B_θ at the core
Reversed Shear	$B_\theta?$	Stabilization of curvature driven modes
Enhanced Reversed Shear and Negative Central Shear Modes	$\nabla P_i, V_\phi$	Shafranov shift

fluctuations. The first determinations of the turbulent particle flux were done under the assumption that there were negligible temperature fluctuations and no poloidal asymmetries in the plasma [97], [180], [186]–[188]. In TEXT, it was shown that these fluxes are consistent with the particle flux determined from power balance [178]. The edge-particle flux determined from probes has also been shown to be consistent with the measurements from HIBP near the edge. The local particle fluxes induced by fluctuations are bursty [189], and it is their time averaged that is compared with the power balance results.

The experimental measurement of electron temperature fluctuations at the plasma edge in several devices [98]–[100], [190], [191] shows that the $\tilde{T}_e - \tilde{n}$ phase is close to zero in the SOL and close to π in the plasma [173]. The value of the phase is then such that the effect of $\tilde{T}_e$ in the calculated particle flux is negligible. From all of these experimental results, the hypothesis of electrostatic turbulence being the main cause of plasma edge transport seems consistent. The main remaining issue is the poloidal asymmetry of the fluxes. Measurements at different poloidal locations are needed to complete these types of experiments.

In RFP's and outside the reversal point, the electrostatic fluctuation-induced particle flux is large and dominates

over the magnetic-induced component of the particle flux [192].

- At the plasma core, most diagnostics measure only density fluctuations. The HIBP could measure both $\tilde{n}$ and $\tilde{\Phi}$. However, after many years of operation, the flux information from HIBP is scarce [20], [83] because the near adiabaticity of the core fluctuations makes the determination of the relative phase difficult.

There are many experimental measurements that indicate correlation between fluctuation level and transport [20]. However, in the absence of phase measurement, it is not possible to reach the same conclusion for the plasma core as for the edge, particularly when a few experiments have shown anticorrelation between fluctuation level and confinement time. In the absence of phase determination, the analysis can be carried a step further by using theoretical input for the fluctuation phases [193], [194]. However, this line of research has not been pursued systematically in the larger devices. There is an extreme situation when the correlation between fluctuation level and transport is unequivocal. Such is the case with the NCS/ERS modes [5]–[7] where anomalous transport is practically eliminated; this is correlated with the nearly total suppression of the broad spectrum of fluctuations [149].

The information on fluctuation-induced energy fluxes is even less complete and mostly restricted to edge plasmas. The main results are as follows.

- In TEXT edge plasmas [20], the convected energy flux from power balance is consistent with the electrostatic fluctuation-induced convective flux and so is the conduction flux. The magnetic fluctuation-induced flux is very small compared with the power balance results;
- At the plasma core, the measurement of core temperature fluctuation [96] has allowed calculation of an upper bound to the second contribution to the electrostatic energy flux (Table IV). An upper bound to the correlation $\langle \tilde{T}_e \tilde{E}_\theta \rangle$ is estimated using the measured $\tilde{T}_e$ and the $\tilde{n}$ and $\langle k_\theta \rangle$ measurements from HIBP. The resulting flux is too small compared with power balance results. However, using $\langle k_\theta \rangle$ from FIR scattering results in a larger value for the flux, and the discrepancy is reduced.

In RFP's, the electrostatic fluctuation-induced energy flux is small [191], and the magnetic-induced component is larger [102]. At the plasma core in MST within the field reversal radius, the magnetic-induced particle flux is consistent with the total energy flux calculated from power balance. However, at the edge, the fluctuation-induced fluxes cannot account for the overall energy transport.

D. Confinement of Alpha Particles and Helium Ash

For a burning fusion plasma, alpha-particle confinement is a critical issue. For most of the time period covered in this paper, the alpha-particle confinement could only be studied either theoretically or through experimental simulation using fast ions. Only since 1994, when deuterium–tritium operation started in TFTR, it has been possible to carry out direct experimental studies on alpha confinement [195], [196]. The main results of these studies are as follows.

- Losses shown by scintillator probes in different poloidal positions agree with first-orbit loss models both in magnitude and in pitch-angle distribution [197]. These losses increase with the application of ICRF, but the magnitude of the increase is small.
- Confined alpha-particle distribution in sawtooth-free discharges agree with the predictions of collisional transport [198].

A nonclassical source of α -particle losses is the α -particle-induced instabilities. The α -particle instabilities were a very active area of research even before the TFTR D-T experiments. This research has been characterized by the development of theoretical models of the coupling between fast particles and Alfvén waves, and by detailed experimental simulations of these instabilities [199]–[201]. Hot particles can drive Alfvén waves unstable. The Alfvén waves that are easier to destabilize correspond with frequencies within gaps in the continuum spectrum. These gaps are created by toroidal coupling [toroidal Alfvén eigenmodes (TAE)] [202], elliptical coupling [ellipticity-induced Alfvén eigenmodes (EAE)] [203], finite beta effects [beta-induced Alfvén eigenmodes (BAE)] [204], etc. For very low magnetic shear, unusual in a tokamak

plasma, global Alfvén eigenmodes (GAE) instabilities are also possible. The TAE have been found to be the main relevant instabilities for a tokamak and they have been studied in the most detail. There is good agreement between the theoretical predictions of the TAE instability threshold and the results from simulation experiments with hot ions [205]. The experimental studies were carried out with careful diagnosis of the radial structure of the fluctuations [200]. The calculated structure of the eigenfunctions correspond well with the experimental measurements. The α -particle-driven TAE has now been identified in the recent TFTR D-T experiments [206].

Experiments in the low-shear stellarator W7-AS have shown the existence of a beam-driven mode which corresponds well with the GAE instability [207]. It is another example in which the comparison between different magnetic configurations are very useful in testing some of the basic features of the theoretical predictions.

For an effective power-plant operation, confining alpha particles and removing helium ash are equally essential. This second issue is particularly critical in the improved confinement regimes. In general, it is desirable to maximize the energy-confinement time while at the same time allowing for an effective particle removal. This must be done at all levels, removal from the core, from the scrape-off layer, and from the divertor. From reactor operation, a quantitative criterion has been derived, it requires $\tau_{\text{He}}^*/\tau_E \leq 7\text{--}15$. Here, $\tau_{\text{He}}^* \equiv \tau_{\text{He}}/\varepsilon_{\text{He}}$, where τ_{He}^* is the helium particle-confinement time and ε_{He} is the exhaust efficiency of helium (ratio of helium exhaust rate to the helium integrated core flux).

Helium transport has been studied in many tokamaks [208]. Realistic experimental studies require not only reactor grade plasmas, but also an efficient exhaust system, like the combination of a divertor and a cryopump. Otherwise, it is difficult to extrapolate to reactor conditions. In L-mode plasmas, it has been shown that ratios in the range of $10 \leq \tau_{\text{He}}^*/\tau_E \leq 30$ can be obtained [209]–[211]. This result makes the L-mode an acceptable operation mode from the particle exhaust perspective; however, the energy confinement in L-mode may be too low. Only in recent years have the helium transport studies been carried out in enhanced confinement regimes such as the H-mode with edge-localized modes (ELM's) and the VH-mode in DIII-D [212]. In these plasmas, effective helium exhaust ($\tau_{\text{He}}^*/\tau_E \approx 11$) has been achieved. The same type of results have been obtained in supershots [213] and in I-mode [214]. The supershot regime studies have been extended to deuterium–tritium operation [215].

The He transport studies give hope for an effective ash exhaust from the plasma core in a reactor. To complete the extrapolation of these results to a reactor regime, it is necessary to find the empirical scaling of $\tau_{\text{He}}^*/\tau_E$ with ρ_* and other dimensionless parameters. The problem of the effective removal of helium at the plasma edge and SOL is still an open issue.

E. Transport and Fluctuations in Stellarators and RFP's

The transport studies discussed thus far are mostly done in tokamaks. Because of the different level of investment in

stellarators and RFP experiments, it is difficult to put the results of these three configurations on the same footing. Therefore, it is necessary to make some separate comments about stellarators and RFP's.

The present generation of stellarator devices [57], [216], [217] is characterized by an average plasma-minor radius of about 20 cm. In this size plasma, transport studies can be carried out without being dominated by edge-plasma effects [26], [218]. They operate with zero current and mostly ECH and NBI heating. The energy-confinement time in these stellarators is about the same as in tokamaks of the same minor radius. However, there are some differences in the underlying transport mechanism that lead to somewhat different transport scaling laws and properties of the equilibrium profiles.

Most of the stellarator discharges have an L-mode-type behavior. The global transport has been described by the so-called LHD scaling [219]

$$\tau_E = 0.17 P_{\text{tot}}^{-0.58} n^{0.69} B^{0.84} a^{2.0} R^{0.75} \quad (19)$$

where the same normalization conventions as in (9) and (10) have been followed. There is a remarkable similarity between this scaling law and the tokamak L-mode scaling law, (10), with two important differences. First, (19) has a positive dependence with density that tends to offset the deterioration of confinement with power. However, there is evidence that this density dependence could be a low-density effect, like the Alcator scaling for tokamaks, and confinement saturates as a function of density [220]. Second, tokamak confinement scaling depends on plasma current and stellarator (in zero-current operation) on magnetic field. This probably means that the real parameter controlling the L-mode confinement is the poloidal field as is inferred in [221].

In stellarators, the local-transport scaling seems to be closer to gyroBohm than to Bohm, as has been shown in the Advanced Toroidal Facility (ATF) [222], the Compact Helical System (CHS) [223], and the Wendelstein VII-Advanced Stellarator (WVII-AS) [26]. This result could be due to the dominance of the electron-loss channels in present-day stellarators. Another characteristic property of the stellarator transport is the lack of the electron temperature profile resilience that has been observed in tokamaks. In WVII-AS and Heliotron-E, it has been shown that the electron temperature profile depends on the heat deposition profile.

In WVII-AS, H-mode-type discharges have been obtained [224], [225]. The H-mode has all the characteristic properties of the tokamak H-mode including the edge-transport barrier, the increase of poloidal flow, and the reduction of edge turbulence. The confinement improvement is still small when compared with the tokamak H-mode, about a 1.3 enhancement factor, and it has a peculiarly narrow operational range in rotational transform. A similar H-mode phase was also observed in the CHS stellarator [226] and very recently in the H-1 heliac [227]. In stellarators, the possibility of external control of the magnetic field opens many opportunities for transport studies. This property has proven useful in testing some basic consequences of neoclassical theory, like the bootstrap current. Because of the variety of trapped particles, the external control

of the magnetic field is an important tool for trapped-particle instability studies that have seldom been used [196].

There is not yet much evidence for internal transport barrier formation in stellarators. However, the multiple-root solution to the radial electric field equation opens new possibilities which are not available to the tokamak. A jump between the electron and ion roots can create a strong radial electric field shear and act as a transport barrier. This seems to be the case in the W7-AS, where the creation of an electron-channel barrier has been reported [151].

Other improved confinement modes have been observed in stellarators. In Heliotron E, both a high ion temperature mode and a pellet-enhanced confinement mode have been reported [228], [229]. The confinement increase has been correlated with an increase of the radial electric field over the L-mode.

RFP dynamics is quite different from a tokamak and stellarator [230]. The RFP equilibrium state is the result of a relaxation to a minimum energy state constraint by helicity conservation, a Taylor state [231]. This state and corresponding profile information is defined by a single parameter, the pinch ratio θ . Resistivity diffuses the current profile away from the Taylor state; this destabilizes a broad spectrum of $m = 1$ tearing modes. The overlap of neighboring magnetic island generates current sheets resonant with $m = 2$ stable modes, causing the transfer of energy at higher m 's and the saturation of turbulence [232], [233]. Tearing-mode activity also modifies the current profile toward the Taylor equilibrium, countering the effect of diffusion. Therefore, at the plasma core, the transport dynamics is governed by tearing modes. The tearing modes convert magnetic energy density in the core poloidal field to magnetic energy density in the toroidal field. Hence, these fluctuations play the dual role of generating the dynamo and inducing anomalous transport. In the experiment, broadband magnetic fluctuations have been measured [234] and they seem to correlate with transport. As discussed before, systematic studies of fluctuation-induced flux have been carried out in MST. The core flux within the field-reversal radius is large and is consistent with the heat flux determined by power balance. However, at the plasma edge, the magnetic-induced flux is low [102]. In the edge region outside the field reversal radius, the total energy flux is convective [235].

Confinement in RFP's has been characterized by a positive dependence of the energy confinement time on the plasma current. Linear and fractional power dependences have been proposed and present experimental data cannot yet resolve this dependence. There is also evidence of the existence of confinement improvement regimes in RFP's. The induction of a poloidal current in MST causes a reduction of the fluctuation level and an increase in confinement time [236]. An increase of up to a factor of five has been measured [237]. The trade-off between confinement improvement and maintaining the dynamo action is one of the critical issues in the optimization of the RFP performance.

An important feature of the edge-turbulence studies is that comparisons between different magnetic confinement devices (tokamaks, stellarators, and RFP's) have been made. The edge plasmas in these devices show similar characteristic properties [238], namely nonadiabatic fluctuations, $\tilde{n}/n \geq \tilde{T}_e/T_e$,

$k_{\perp}\rho_s \approx 0.1$, and $\tilde{n}/n \approx (k_{\perp}L_n)^{-1}$. The bursty nature of edge-particle fluxes have been observed both in stellarators [189] and tokamaks [239]. These fluxes have very similar probability distributions for both types of devices. Since all of these devices have rather different edge-magnetic configuration, these comparisons are very important in identifying the basic turbulence mechanism at the plasma edge.

V. PROGRESS IN UNDERSTANDING ANOMALOUS TRANSPORT

Understanding the transport mechanisms requires theoretical formulation of those mechanisms and their accurate experimental testing. To validate a theoretical model, experimental tests must be done at two levels: fluctuation dynamics and transport. For many years, theory has identified possible mechanisms underlying turbulence in magnetically confined plasmas (instability drives and linear theory of these instabilities), and from them developed expressions for the turbulent diffusivities [29], [30]. Theoretical determination of turbulent diffusivities is far from straightforward, but it is unavoidable in testing transport theories. The complexity of the plasma physics experiments and the strong nonlinear couplings between the observables make the experimental tests of theory very difficult and sometimes obscure. A full solution of the transport problem would imply understanding plasma turbulence and, while great progress has been made, we are still far from this point. However, the lack of a complete transport theory does not imply a total lack of understanding of the transport mechanisms. The challenge is to construct a transport theory from the fundamental mechanisms and to formulate the right experimental tests. This challenge has been met in a few areas where real progress on understanding has been made. A few of those areas have been selected for discussion.

A. Turbulence Understanding in Magnetically Confined Plasmas

As previously discussed, turbulence is ubiquitous in magnetically confined plasmas. Fluctuation measurements at the core or at the plasma edge show different levels of fluctuations, but rarely is the plasma quiescent. Turbulence is believed to be the main cause of the anomalous transport. Independent of the practical goal of understanding transport, it is a scientific goal to understand turbulence. In the last 25 years, clear advances have been made toward that challenging goal.

The basis for a turbulence theory was set in the mid 1960's mainly by pioneering work of the USSR school [240], [241] and Dupree [242], [243] in the United States. The limits of strong and weak turbulence were investigated. Saturation mechanisms for plasma instabilities such as mixing length and resonance broadening were proposed. In the 1970's, the turbulence studies focused mostly on drift-wave turbulence. The basis of the turbulence theory was consolidated with the work of Dupree and collaborators [244]. This work was the basis for one of the first theoretical anomalous transport models [245] trying to explain Alcator scaling. The underlying instabilities for this model are the dissipative drift waves, which are linearly stable in toroidal geometry, but destabilized

by the nonlinear diffusivity. At the same time, renormalization theory for drift-waves turbulence at kinetic level, including Compton scattering, was developed [246], [247]. This work led to theoretical predictions for the spectral decay index and broadening of the frequency spectrum at a fixed k .

To study drift-wave turbulence, the Hasegawa–Mima equation [248] became the standard model. In its derivation, the electron response is assumed to be adiabatic, therefore, it only contains the polarization drift nonlinearity. The Hasegawa–Mima model predicts an inverse cascade of total energy, from large $k_{\perp}$ (small radial scales) to small $k_{\perp}$ (large radial scales). The adiabatic assumption excludes the density dynamics and makes this model rather limited. To break the adiabaticity, the so-called $i\delta$ model was introduced [249]. This model invokes an $i\delta\tilde{\phi}$ nonadiabatic component to the density fluctuations. A consequence of the nonadiabatic contribution, the model includes a second nonlinearity—the $E \times B$ drift nonlinearity—convection of nonadiabatic electrons. Analytical and numerical calculations [250], [251] have treated different $k_{\perp}\rho_s$ dynamical ranges. Since the two nonlinearities dominate in different parts of the spectrum, a full self-consistent model is required for studying the spectral energy transfer. When that is taken into consideration, it has been shown that the $E \times B$ nonlinearity transfers energy (nonlocally in k -space) from large to small scales while the polarization drift transfers energy locally from small to large scales. For $k_{\perp}\rho_s \ll 1$, the $E \times B$ nonlinearity dominates and the spectrum is flat. For $k_{\perp}\rho_s \sim 1$, both nonlinearities must be taken into account, with the $E \times B$ nonlinearity blocking the condensation effect to long scales [252], [253].

These simple drift-wave models have produced clear qualitative predictions for the dynamics of fluctuations. The question is how to test these dynamics in the experiment. Analysis methods for fluctuation data have been evolving toward extracting information on the nature of the nonlinear couplings. For quadratic nonlinearities, the use of bispectral analysis techniques [254] coupled to a simple model can lead to the evaluation of the spectral transfer [87], [255]. Because some of the present-time data variations complicate the interpretation, an extension of these techniques to wavelet analysis has been proposed [256], [257] and applied to experimental measurements of plasma-edge fluctuations [173].

Many basic plasma turbulence experimental studies have been carried out in simple geometry machines, like Q-machines. However, the number of such experiments has greatly decreased in recent years while theory was improving its level of understanding. It is a pity not to have such experiments to check the more sophisticated aspects of present theoretical formulations.

Another fundamental aspect of the turbulence is its possible intermittent or “granular” character. This reflects the underlying dynamics and affects the closure assumptions in the renormalization of the model equations. Some turbulence models have led to the existence of long-living structures [258], [259] even when magnetic shear is taken into account [260]. Numerical studies contrasting the presence and absence of long-living structures have been done to give a characterization of statistical quantities such as skewness and kurtosis,

that allow a comparison with experiment [261]. Most experimental results are consistent with Gaussian statistics within the error bars of the experiment (which are not small enough to exclude the existence of such long-living structures) [262]. To further explore the statistical nature of the fluctuations, it is necessary to apply more sophisticated analysis techniques, like conditional statistical analysis [263].

Fluid-like turbulence models (resistive MHD instabilities, η_i instabilities well above the threshold, etc.) have also been studied analytically as well as numerically. In his pioneering work on plasma turbulence, Kadomtsev [240] applied the mixing length approach to fluid-like plasma turbulence models. The mixing length Δ is defined as the mean transverse displacement of fluid elements. In the case of a passive scalar, like density n , the density fluctuation level is essentially given by $\tilde{n} \approx \Delta dn/dr$. Since the fluctuating velocity $\tilde{V}$ can be related to the density fluctuation by $\gamma \tilde{n} \approx \tilde{V} dn/dr$, the induced particle flux is

$$\langle \tilde{V} \tilde{n} \rangle \approx \gamma \Delta^2 \frac{dn}{dr}. \quad (20)$$

Here, γ is the linear growth rate of the underlying instability. Equation (20) implies that the turbulence-induced transport is essentially diffusive with an effective diffusion coefficient $D_{\text{eff}} \approx \gamma \Delta^2$. When density fluctuation measurements from many experiments are put together, they show a basic scaling consistent with the mixing length argument [18], [264].

The renormalization approach has also been applied to fluid-like instabilities. It proceeds through perturbation theory with a recursive determination of the nonlinear time scale τ_c . If this time scale is approximated by the linear time γ^{-1} , conservation laws are violated. In some simple models, the renormalization of the nonlinear equations can be treated analytically. For instance, for pressure-gradient-driven turbulence (resistive interchange, resistive ballooning modes, neoclassical pressure-driven current-diffusive interchange instability, etc.), with a pressure evolution equation given by simple convection, the renormalized pressure evolution equation for the long radial scales is essentially

$$\frac{\partial p}{\partial t} = -\tilde{V}_r \frac{dp_0}{dr} + D_{\perp} \nabla_{\perp}^2 p \quad (21)$$

where the first term in the right-hand side (rhs) is the pressure convection and the second term is the turbulence-induced diffusion with $D_{\perp}$ the amplitude-dependent turbulent diffusivity. By balancing the diffusion term with the time-evolution term, we recover the mixing length result. However, once we have a dynamical equation, we can calculate the true saturation of the turbulence by setting $\partial p / \partial t = 0$ and solving (21) with the turbulence dynamic equations as an eigenvalue problem for $D_{\perp}$. This leads to solution of the form [265]–[267]

$$D_{\perp} = \gamma \Delta^2 \Lambda^2 \quad (22)$$

where Λ is a slow-varying function (constant or logarithmic) of the other dimensionless parameters in the problem. Therefore, we recover a modified form of the mixing length prescription. However, in (22), there are no ambiguities in the meaning of γ and Δ .

The scale invariance approach [165], [268] is an alternative approach for the determination of the turbulence-induced diffusivities. It is based on the idea of intermediate asymptotics [269]. In this approach, one identifies all the independent transformations under which the dynamical equations are invariant. When applied to the same turbulence models as discussed above, an expression of the same form as (22) is reached [268]. However, with this method, the function Λ remains undetermined. This approach has been applied to several types of turbulence dynamics and it can be a powerful tool in identifying the main dependences of the diffusivities. The minimal form of the scale invariance approach is the dimensional analysis.

Both approaches, renormalization of the dynamic equations and scale invariance, have also been applied to electron thermal instabilities (rippling modes, drift thermal instabilities, thermal drift waves, etc.). In this case, the saturation mechanism is different from the modified mixing length mechanism described above. The saturation is caused by the balance of the instability drive with enhanced parallel diffusion by radial turbulent transport balances [270]. This mechanism is the equivalent to the resonance-broadening saturation of Dupree, but in real space.

The existence of saturation mechanisms other than mixing length should caution on its indiscriminate use. Numerical calculations should be used to test the validity of the saturation mechanism and its regime of application. If subcritical transport mechanisms can operate in magnetically confined plasmas, the mixing length approach can lead to totally misleading results; this is a particular concern for models that are dominated by near-marginality dynamics.

As the computer capabilities increased and the fluid models became capable of incorporating Landau damping, turbulence calculations have moved closer to the simulation of the experiment. However, it is still necessary to continue the studies with simpler models to reach an understanding of the fundamental turbulence mechanisms.

B. Understanding Ion-Channel Anomalous Transport Mechanisms

In the late 1960's and early 1970's, the basic instabilities linked to the ion temperature gradient were identified [271]. The so-called η_i mode (where $\eta_i \equiv [d \ln(T_i)/dr]/[d \ln(n_i)/dr]$) is an electrostatic sound wave driven unstable by the ion-pressure gradient. This instability is triggered by η_i becoming greater than a threshold value, η_{th} . The η_i has a variety of branches in different regimes. At low collisionality, the temperature gradient destabilizes trapped-ion modes which were most feared because of the large radial-scale lengths associated with these instabilities [272]. Although the basic instabilities were identified in simple magnetic geometry, the two-dimensional structure of the eigenfunctions [273], [274] and the different nonlinear mechanisms for saturation [275], [276] of the instabilities were not addressed until the mid 1980's. Two facts slow down research of these instabilities: the tokamak ohmic experiments suggested that the electron channel was the dominant loss

channel and the PLT experiments did not show any of the deleterious effects expected from the trapped-particle modes. Later on, when the Alcator C pellet experiments showed that the peaking of the density profile (reduction of η_i) brought an improvement in confinement, the study of η_i instabilities gathered momentum. Also, the accurate measurement of the ion-temperature profiles [72] showed that $\chi_i > \chi_e$ in L-mode plasmas brought attention to the ion-channel instabilities. In recent years, a great deal of theoretical and computational activity has been concentrated on ion-temperature gradient-driven turbulence (ITGDT) models. The experimental evaluation of the relevance of η_i models has been oscillating [277], [278]. The first analytical results for χ_i were derived for η_i well above the stability threshold where a fluid approximation could be applied. The value for χ_i obtained in these calculations was too high compared with the experiment. The fact that η_i instabilities can lead to such high levels of losses suggests that a marginal stability condition could constrain the profiles [279]. The determination of the marginal stability criterion and the linear eigenmode close to the threshold requires the use of kinetic theory and full-toroidal geometry. Analytical work has progressed in refining the calculation of stability thresholds [280]–[282].

The combined difficulties of including kinetic effects and realistic geometry make the analytical nonlinear problem close to intractable. There are multiple regimes and diverse forms of the instability that have to be pieced together in constructing a turbulent diffusivity based on the ion-temperature gradient-driven instabilities. All these effects combined make it very difficult to do comparisons with experimental results. When experimental profiles were used to evaluate the theoretical χ_i for comparison with the measured χ_i , the results were negative [283]. However, because of the strong nonlinear dependence on the profiles, measurement uncertainties are highly amplified. It is more reasonable to use the theoretical χ_i to make a prediction of the profiles and do a comparison for the self-consistent solution. Transport models have been built on the basis of the mixing length approach and using ITGDT as the basic instability. These models must be complemented with some effective plasma-edge transport [284], [285]. They had been relatively successful in simulating experiments. The analytic multimode approach has been taken to a high degree of effectiveness [286], [287] combining more detailed analytic versions of ITGDT [288] for the plasma core and resistive ballooning-induced transport losses at the plasma edge [289], [290].

A new opportunity occurred with the development of gyrofluid models with Landau closure [291]. The possibility of including Landau damping (in the linear approximation) in fluid models has extended the computational capabilities for studying ITGDT [292], [293]. These techniques led to a new approach for calculating the anomalous diffusivities [294]. Following this approach, the turbulent diffusivities are expressed in the same form as (22), $\chi = W_{nl} \langle \gamma / k_{\perp}^2 \rangle$, where the main dependences come through the mixing length contribution $\langle \gamma / k_{\perp}^2 \rangle$ and W_{nl} is the slow-varying function of the relevant dimensionless parameters. Therefore, γ and $k_{\perp}$ can be accurately calculated; fast linear gyrofluid codes can now

do so in an efficient manner. The coefficient W_{nl} is weakly dependent on the dimensionless variables, as has been shown in the few analytical calculations of turbulent diffusivities in fluid regimes [295]. Therefore, W_{nl} does not need to be so accurately calculated. It is represented by a simple parameterization and the parameters are calculated with the nonlinear gyrofluid turbulence code. Transport calculations can use this simple parameterization to evaluate W_{nl} when it is needed. This approach [296], [297] allows fast calculations without the usual analytic restrictions on collisionality regimes, values of T_i/T_e ratios, etc. The calculated diffusivities can be coupled to a transport code. These calculations have confirmed that the ITGDT is dominant at the plasma core in L-mode plasmas. Transport calculations following this approach give a good description of TFTR confinement results [293].

In parallel with the development of gyrofluid calculations, there has been a substantial step forward in particle simulations. One can get a good perspective from the review paper by Dawson [111]. Gyrokinetic calculations [298] have advanced significantly by incorporating full three-dimensional geometry. They are providing reliable information on the basic properties on ITGDT, such as fluctuation levels, spectrum, and induced transport [299], [300]. As noted by Dimits *et al.* [300], the most sophisticated noncollisional fluid simulations use six-fluid-moment gyrofluid models, whose nonlinear validity is not known *a priori*. Since the toroidal ITG mode is a kinetic instability, the demonstration of the validity of the fluid representations of the nonlinearities is important. Comparisons between fluid and particle approaches are a major goal in the near future.

The initial gyrofluid and gyrokinetic models included only ion dynamics. Electron dynamics is being now included in some numerical calculations [301], [302], although they are still being applied to tokamak regimes with dominant ion dynamics.

The ITGDT picture of core dynamics has also been tested from the point of view of the fluctuations. Gyrokinetic simulation of ITGDT [303] showed k_{θ} and k_r spectra that agrees well with the one measured with beam-emission spectroscopy in TFTR [87].

The ITER expert group on transport modeling is carrying out the validation of a number of transport models by comparing their predictions with experimental data from a range of tokamaks in the ITER profile database [304]. This group has defined a set of measures of performance that allow quantitative comparison of the performance of the different models. Using experimental information for the edge boundary conditions, most of the theory-based models give a good description of the data (from 20–30% deviations). This represents a considerable improvement over the situation just a decade ago and raises hope for effective predictive modeling in a near future.

All of these results indicated that the ion-core plasma transport dynamics seem to be well described by the traditional model based on ITG turbulence. However, a complete model does not yet exist. The detailed formulation of the diffusivities is now being worked out. It is of fundamental importance to test if the inclusion of linear Landau damping in the

gyrofluid calculations is good enough. In wave propagation, that is not the case and in the asymptotic states nonlinear Landau damping is essential [305]. The inclusion of shear-flow effects and electron dynamics is also essential. Detailed comparison between kinetic and fluid calculations are needed to validate these models. Also, a more detailed study of plasma core fluctuation dynamics with comparison with nonlinear simulations is needed to set on a solid basis the understanding of the basic transport mechanism.

Since many of the transport models based on ITGDT are stiff, their predictive capabilities strongly depend on the assumed boundary values. This shifts the problem of understanding confinement to understanding transport at the plasma edge. Regrettably, the edge transport mechanism still remains an open question.

C. Understanding Fluctuation Suppression by Electric Field Shear

The universality of the H-mode suggested the existence of a generic mechanism that explains the reduction of the fluctuation level and formation of a transport barrier. After the causal relation between the radial electric field and the L to H transition was first suggested [306], it was followed by more detailed L to H transition models [307] based on a two-step process: 1) at a certain level of injected power in a tokamak, the edge-particle losses create or modify the radial electric field, and 2) this electric field is then responsible for the reduction of the fluctuation level and transport. These models did not consider in detail the dynamics of edge-turbulence suppression. A general scaling model [308] was proposed to explain the effect of shear $E \times B$ flow on turbulence. The scaling model showed that, in the presence of an $E \times B$ shear flow, the turbulence decorrelation time τ_c is a hybrid time scale of the shearing time scale $\tau_s \equiv 1/\omega_s = 1/(k_\theta V'_\theta \Delta)$, and the diffusive time scale, $\tau_D \equiv \Delta^2/D$. Here, Δ is a characteristic radial scale of the turbulence, D is the turbulent diffusivity, $V_\theta \approx -E_r/B$, and E_r is the radial electric field. Therefore, the decorrelation time is $\tau_c = 1/(k_\theta^2 V_\theta'^2 D)^{1/3} = 1/\tau_D^{1/3} \tau_s^{2/3}$. When the shearing effect dominates, $\tau_s < \tau_D$, the effective turbulence decorrelation time is reduced. As a consequence, fluctuation levels should be reduced. To apply this criterion to a toroidal system, the form of the shearing rate ω_s had to be generalized [309] to toroidal geometry

$$\omega_s = \frac{\Delta_\rho}{R\delta\varphi} R \frac{\partial}{\partial \rho} \left(\frac{E_\rho}{RB_\theta} \right) \quad (23)$$

where

- ρ radius-like flux surface label;
- Δ_ρ radial correlation length;
- $R\delta\varphi$ toroidal correlation length of the fluctuations.

Note that a change in magnetic shear also influences the shearing rate through B_θ . Another associated effect of $E \times B$ shear decorrelation is the change of the relative phase between electrostatic potential and the other fluctuating quantities, like density and temperature. This change in the relative phase causes a reduction in the fluxes even when the fluctuation level is not changed [310], [311]. How strong the change of

phase is depends on the dissipation associated with the particular fluctuating quantity. This different change of phase can cause different levels of suppression in the particle, electron, and ion-energy channels. Turbulent suppression by $E \times B$ shear flow has been confirmed in the numerical calculations of ion-temperature gradient-driven turbulence [312], [313], high- k drift-resistive ballooning turbulence [290], drift-wave turbulence [260], drift-thermal turbulence [314], and resistive pressure-gradient-driven turbulence [315], [316]. An example of fluctuation suppression by a self-consistent sheared poloidal flow is shown in Fig. 6.

The basic concept of the $E \times B$ shear decorrelation of the turbulence has also been tested experimentally and the results support this concept. The first experimental evidence was from the plasma edge in TEXT tokamak [317]. At the velocity shear-layer position, the decorrelation time of the turbulence is shorter than in neighboring regions. One of the first experiments to measure fluctuation suppression was at the L-H transition [73], [93]. In DIII-D and with reflectometry measurements, it was shown that the fluctuation suppression region coincides in time and location with the increase of radial electric field and the formation of a transport barrier (Fig. 3). More detailed evidence was obtained with probe measurements [318]–[321] and FIR scattering [322]. In the case of probes, measurements suggest that the transport reduction is presented in two phases: reduction of flux by phase decorrelation and later suppression of fluctuations. In CCT, a poloidal probe array indicated that the fluctuations had a strong ballooning-like asymmetry before the transition. The remaining fluctuations in the H-mode were poloidally symmetric [323].

Fluctuation reduction by $E \times B$ shear flows has also been studied in experiments with an externally imposed radial electric field. There are multiple ways of biasing the plasma edge [324] depending on the edge geometry. With this technique, an H-mode-like confinement regime can be triggered [122], [123], [125], [226], [325], [326]. These techniques also allow a controlled experiment in which the electric field is varied and its effect on fluctuations can be measured with probes. These types of experiments have been carried out in several devices [319], [327] showing clear correlation between electric field shear and effective reduction of fluctuation levels. It has also been shown that by edge biasing the power threshold required for a spontaneous transition is reduced [319], [326], [328].

The radial profile of the E_r during a spontaneous H-mode has been studied in different devices, like DIII-D [73] and JFT-2M [329]. They all show the same well-shaped structure. The $E \times B$ shear decorrelation mechanism is not only in operation in the H-mode, but it is possibly a generic mechanism for confinement improvement. In the VH-mode, there is a clear correlation between the penetration of the $E \times B$ shear and the region of confinement improvement [120]. To further test this mechanism, an external coil was used to reduce the toroidal velocity [330] (through magnetic braking) and the reduction of the corresponding $E \times B$ shear caused the elimination of the confinement improvement. This mechanism also seems to be in operation in the enhanced reversed shear and negative shear-confinement modes [7], [149].

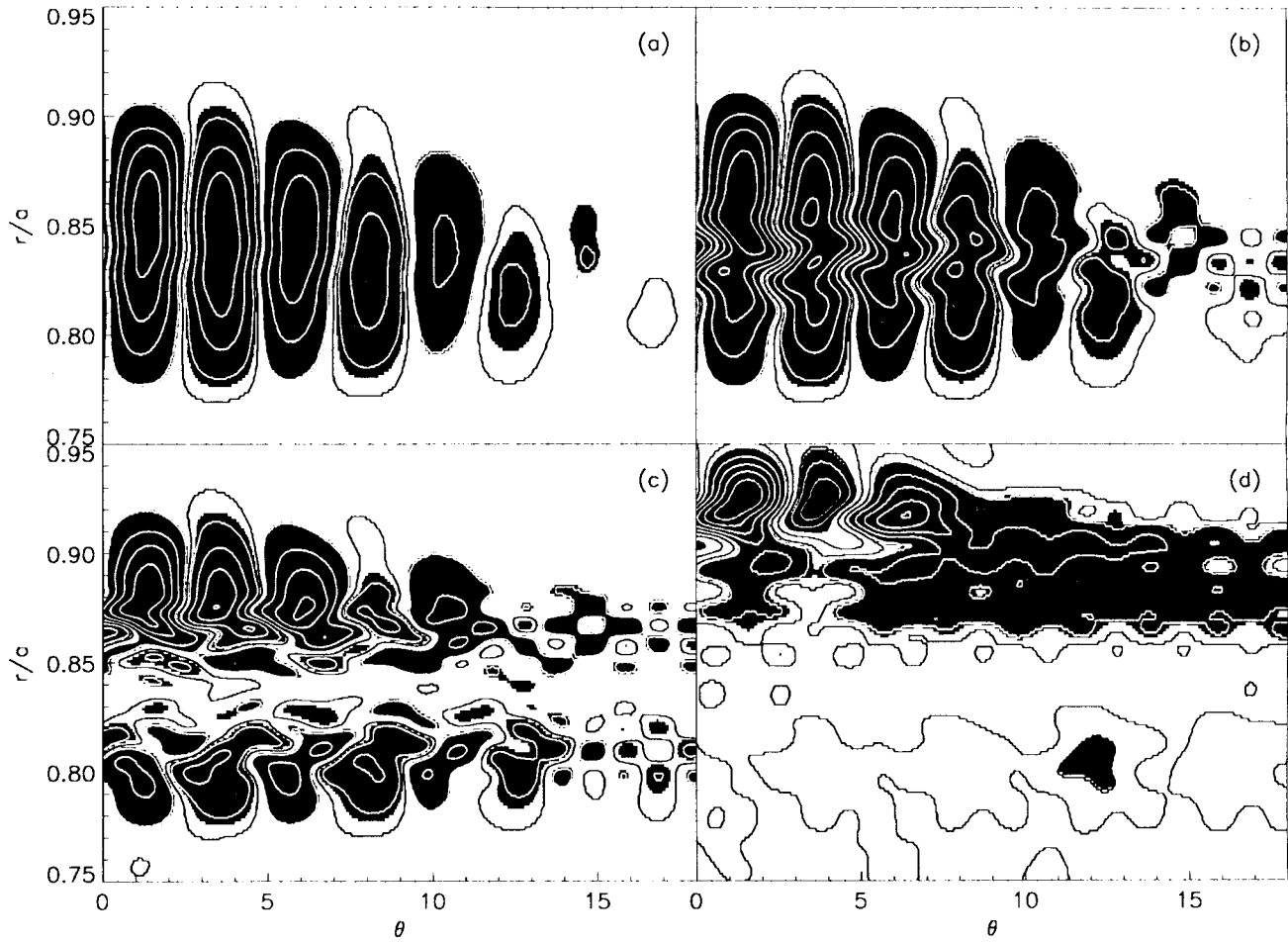


Fig. 6. Fluctuation suppression by $E \times B$ shear flow [461]. Contours of constant density fluctuations at four times on the nonlinear evolution of negative compressibility turbulence: (a) in the growing phase of the instability, the fastest growing dominates; (b) through Reynolds stress, poloidal shear flow is amplified as shown by the distortion of the contours; (c) when the flow is large enough, the eddies are sheared; and finally (d) there is total suppression of the fluctuations.

Less straightforward is the incorporation of this concept in models for improved confinement regime. In many of these confinement regimes, there is clear evidence of a bifurcation and the shear suppression must be coupled to a generation mechanism for the radial electric field that, through a positive feedback loop, causes the transition. There are multiple possible sources for the electric field [331], one already mentioned is the orbit loss. We can identify others by looking at the radial force balance in steady state

$$\frac{\langle E_r \rangle}{B_\varphi} = -\langle V_{\theta i} \rangle + \langle V_{\varphi i} \rangle \frac{B_\theta}{B_\varphi} + \frac{1}{Z_i |e| B_\varphi} \frac{1}{\langle n_i \rangle} \frac{\partial \langle P_i \rangle}{\partial r} \quad (24)$$

where the angular brackets indicate flux surface averaged. Here, we see that an electric field can be generated by a poloidal flow, toroidal flow, and ion-pressure gradient. The shear in the radial electric field can be induced by the shear in the mentioned variables and/or through poloidal magnetic field shear or a density gradient. Depending on the characteristics of the confinement mode, one of these terms or more likely a combination, could be the cause of the $E \times B$ shear. Just in the L- to H-mode transition, several mechanisms have been suggested.

- Orbit loss. Near the plasma edge, ions on banana orbits are lost and, as a result, a radial electric field is created [306], [307], [332]. Bifurcation is caused by the nonlinear dependence of the neoclassical viscosity on the radial electric field [333]–[335].
- ∇P_i is the main electric field contribution, once coupled to $E \times B$ shear suppression, it leads to a bifurcation in the transport equation [336], [337]. The transition is like a first-order critical transition [338], that is, the flux is discontinuous at the critical point.
- Poloidal rotation is the dominant contribution to E_r . It is induced by the poloidal asymmetry in the particle anomalous flux, Stringer spin up [339].
- Poloidal rotation is the dominant contribution to E_r . Plasma turbulence changes the poloidal-flow profile via the Reynolds stress and increases E_r' [316], [340], [341]. This causes a transition to a state dominated by a global poloidal flow with reduced-turbulence fluctuations [342], [343]. This transition is like a second-order critical transition [338], that is, the flux is continuous across the transition, but its derivative is discontinuous.
- Poloidal rotation is the dominant contribution to E_r at the transition, but ∇P_i effects are coupled. Reynolds stress is

the cause of the poloidal-flow profile modification [344], [345]. This model gives two phases in the transition, a slow second-order-type transition, where the poloidal rotation is the dominant term in E_r , followed up by a fast first-order transition (if power is high enough above the threshold). In the second phase, the electric field shear is dominated by ∇P_i . The poloidal velocity decays because there is no longer the turbulence causing the amplification of the shear flow.

When the transition is externally triggered, there are also diversified interpretations on how the electric field is created [324], [346], [347].

There are other transition models that do not directly involve the radial electric field; for instance, an alternative is ω_i^* stabilization effects [348] coupled to the density gradient. Other models are based on the role of the neutral particles at the edge [349]. In comparing all of the transition models with the experiment, the difficulty is on separating cause from effect.

In the transition models, the nonlinear coupling between E_r and profile gradients (including ideal MHD instabilities) is such that fixed-point solution can change to limit cycles or even more complex behavior. This oscillatory behavior may explain the dithering and ELM phenomena [350]–[352].

Is the $E \times B$ shear a universal mechanism that explains most of the improved confinement regimes [353]? There is some evidence that $E \times B$ shear could also be the basic mechanism for the ERS/NS confinement regimes [149] and that theoretical models have been advanced for these new regimes [354]–[356]. Of course, alternate explanations have also been proposed. These alternatives are not mutually exclusive because several mechanisms could be simultaneously at work. For instance, in these discharges, first the reverse magnetic shear stabilization reduces the turbulence and lowers the power threshold for a bifurcation induced by either the radial electric field and/or increases the self-stabilizing effects through the Shafranov shift. The synergism of this multiple mechanism leads to the suppression of turbulence. This synergism could also be present in the L to H transition where the potential role of the magnetic shear has also been raised because of the transitions triggered by rapid-current ramp down [357].

In stellarators, the high ripple enhances the flow damping; therefore, by comparing tokamaks and stellarators, improved confinement modes may be possible to identify the role of different terms in (24). This approach suggested and applied in [230] identifies the pressure-gradient term as a dominant effect in the hot-ion mode.

Possible E_r field sources for the enhanced confinement regimes together with alternate explanations for the confinement improvement are listed in Table IV. A systematic study of these different regimes with evaluation of the electric field and with fluctuation measurements is needed. The different responses of the electron and ion channels to the transport barrier formation is an issue that requires more theoretical attention. The accurate incorporation of $E \times B$ shear in the theoretical transport models discussed in the previous section is also one of the pressing issues in making those models capable of addressing different confinement regimes.

D. Magnetohydrodynamic (MHD) Stability and Operational Space of Experimental Devices

MHD theory was well established in the late 1960's [358], [359]. What we have seen in the last 25 years is the systematic application of the theory to experiment and development of some new ideas. This subject is beyond the scope of this paper, but is brought up here because of the impact it had in the understanding and control of anomalous transport. The impact of MHD understanding has been in:

- the development of linear MHD codes and its application to the interpretation of the experiment has provided an understanding of how to optimize profiles and broaden the operational space of the devices. This has allowed access to regimes of better confinement;
- new ideas such as beta self-stabilization [360], [361] and the concept of second-stability regime [362] not only apply the global MHD instabilities, but also to all curvature-driven microinstabilities. Operation in the second stability regime not only allows one to reach higher beta values but also to operate with improved confinement.

The impact of these ideas on transport studies has been in two areas: 1) pushing toward high-beta poloidal to increase the self-stabilization effect of the Shafranov shift and gain access to the second stability regime and 2) reducing or reversing the shear to stabilize ballooning modes and curvature-driven microinstabilities. The current profile control is the primary tool to explore these possible regions of confinement improvement. These mechanisms can provide an explanation for the high β_p and RS confinement modes [363]–[366].

MHD is the confinement limitation left to many of the enhanced confinement regimes. At present, many of the high-confinement discharges have the confinement reduced and even sometimes terminated by MHD activity. MHD instabilities may control the gradient of the plasma-edge pedestal in H-mode confinement. The ideal ballooning limit has been used in some models to describe the edge gradient [367]. However, in shaped tokamaks, the plasma edge is often in the second-stability regime. Therefore, some other mechanism, such as the edge peeling modes [368], may be the ultimate mechanism controlling the pedestal gradient.

Resistive MHD instabilities that were present in many of the relatively low-temperature tokamaks in the 1970's are less likely to affect operation in high-confinement discharges. Below the ideal MHD instability threshold, the observed low-toroidal mode number instabilities are in some cases well described by neoclassical pressure-gradient-driven instabilities [369]. If these types of instabilities are active, they could strongly affect the transport properties of the plasma. In any case, MHD stability optimization must be developed in parallel with the confinement optimization.

VI. OPEN ISSUES

There are still some open issues relative to understanding transport. In the previous sections, we have concentrated on the topics in which an emerging understanding exists. However, there are other topics for which conflicting results make it

difficult to converge on a clear picture. A few of these outstanding issues in our transport understanding are reviewed in this section. The first three issues directly challenge our understanding of ion transport, although they do not invalidate it. There are potential solutions within the present theoretical framework, but they are not simple.

A. Bohm Scaling of L-Mode Confinement

Our understanding of the core transport in the ion channel is based on theoretical models that lead to gyroBohm-type scaling. Core fluctuation measurements also give radial correlation length for the fluctuations, which are very short compared with the device minor radius and possibly compatible with the core-dynamical models. However, the scaling derived for the ion-thermal diffusivity on the bases of dimensionally similar discharges is Bohm or worse than Bohm. How can we reconcile these results? There is no conclusive answer yet, but there are several ways to approach this dilemma.

- Long correlation lengths. There are possible ways of having long correlation lengths for the turbulence. One is by having long radial scale modes, like trapped-ion modes [254] or basically through memory of the linear toroidal coupling [370]–[372]. This implies correlation lengths of the order of the magnetic shear parameter. The measured correlation lengths of turbulence are relatively short. However, the turbulent structures associated with these instabilities are not strictly radial. Therefore, in calculating the correlation length of the turbulence, 2-D fluctuation measurements are needed.
- Fast propagation. Because of the toroidal coupling, information between different radial position could be transmitted rapidly. A nonlinear computational model has been developed to study this mechanism. This model shows that fast propagation is possible, but it is in a restricted parameter regime [373].
- Dynamics near marginal stability [374]–[376]. Some of the experimentally determined profiles are close to marginal stability. Models based on critical gradient scale length, like ITGDT, are clear candidates to present near-marginal dynamics. However, the dynamics near marginal point could be a generic feature of magnetic-confined plasmas, independent of a particular underlying instability. This suggested that plasma equilibrium tended to remain on a self-organized critical state [377]. In such a state, transport length and time scales of all sizes are possible, while fluctuations remain on local scales. This model [378] may also explain some of the puzzles presented by perturbative transport studies as we will see later.
- Modification of the gyroBohm scaling by shear flow [263], [325]. The mixing length is taken to be proportional to ρ_s but the linear growth rate is modified by shear-flow stabilization. The scaling of the decorrelation time of the turbulence with ρ^* could be a test of this model. The incorporation of shear flows to the linear and nonlinear formulation of ballooning instabilities [313], [371], [379], [380] presents serious theoretical and computation issues that need to be addressed.

- A multimode transport model, based on ITG plus edge-resistive ballooning, gives an apparent worse than Bohm ρ^* scaling [381]. The increase of resistive ballooning at low magnetic field causes a steepening of the temperature gradient at the core that worsens the ITGDT contribution.

Dimensionally similar discharges have provided the empirical basis for the Bohm scaling. It is important that such experiments are done with core fluctuation measurements to determine the scaling of the radial correlation length of the fluctuations. These measurements would be very helpful in identifying the dynamic mechanism responsible for the Bohm scaling.

B. Fast Time Scales and Nondiffusive Transport Phenomena

In recent years, experimental evidence has been accumulating on transport phenomena that show propagation time scales significantly faster than diffusive propagation. A few examples can be cited.

- Fast propagation of the confinement improvement caused by the L to H transition in JET plasmas [382]. The experimental measurements indicate that the change in core confinement following the transition propagates inward at a speed larger than 100 m/s.
- In the W7-AS stellarator [383], the experimental electron energy transport coefficients calculated during the transient phase react in about 1 ms to the change in heating power. This propagation is considerably faster than the T_e profile change.
- Ballistic propagation of heat pulses in TFTR, where there is evidence of a very rapid energy deposition well beyond the usual mixing radius of the sawtooth [384].
- Experiments in the TEXT tokamak show that a cold pulse at the plasma edge propagates inward in a time scale much faster than the diffusion time. As it moves inward, the pulse shrinks and disappears near half plasma radius. At the plasma center the temperature rises promptly [106]. Similar experiments have been carried out in TFTR [385] and JET.

Analysis of these experiments, based on diffusivities with “reasonable” nonlinear dependences on T and ∇T , have not been able to explain the fast time scales. Some of these examples have been used as an indication of nondiffusive effects and even nonlocal effects in transport dynamics.

Experimental conditions could be such that the plasma is near a “critical point” (either a transition or a marginal stability). Under such conditions, the model used to interpret the experiment should include the dynamics of the critical point. If the transport model does not include transition dynamics, one can hardly expect to explain the propagation of the effects of the transition. In the case of confinement transition, the structure of transport equations when nonlinearities are included is similar to a reaction–diffusion equation. These equations have front propagation solutions with time scales that are a hybrid scale of the diffusive propagation and the transition time scale. Models of the confinement propagation that include the transition dynamics lead to fast propagation of confinement improvement fronts [386].

In tokamaks, there is also evidence for the resilience of temperature [387], [388] and density profiles; that is, these profiles are not affected by changes in the source profiles. Some experiments suggest the existence of anomalous particle and heat pinches [389]. The situation is not the same in stellarators [26], where the temperature profile follows the heat deposition profile. For tokamaks, observations like the ones just described have been around for many years. These observations led to the proposal of the profile consistency principle [390], and of the existence of canonical profiles [391], [392]. These ideas have been adopted in many transport models and in a variety of forms [284], [393]. Transport models that include a critical gradient scale length with strong transport that forces near marginal operation may explain many of these general features. Such is the case of the IFS/PPPL transport model [293]. More experimental tests of these models using perturbative transport techniques are highly desirable.

The issue is whether the profile consistency is the consequence of the transport dynamics or just the result of sawtooth oscillations at the plasma-core large-edge transport. These two effects pin down the profile at the $q = 1$ surface and at the edge, leaving little freedom for the transport to modify profiles in between.

The use of invariants to describe turbulence [394] in tokamaks offers another alternative. The assumption is that turbulent plasmas in the absence of sources are close to turbulent equipartition. This leads to a connection between density and q profiles. This approach offers possible explanations for the anomalous pinch and energy exchange in tokamaks [395], [396].

Some alternative explanations involve transport mechanisms dominated by magnetic perturbations (we discuss this issue below). The nonlinear growth and interaction of neoclassical magnetic islands produce regions of stochastic field lines. Across these regions fast propagation is possible [397], [398].

The issue of fast time scales and possible critical gradients may be also connected to the Bohm scaling just discussed. Some of the potential solutions listed there involving either long correlation lengths and/or fast propagation could be applied here. It would be desirable to have more systematic experimental evidence and to find a theoretical framework within which these diverse phenomena could be interpreted.

C. Isotope Scaling

From the late 1970's, it is known that confinement in deuterium plasmas is better than in hydrogen [399]. In most confinement regimes, confinement time increases as the square root of the atomic mass, $\tau_E \propto M_i^{1/2}$. The positive scaling with the ion mass is a systematic result observed in most tokamaks and for particle, energy, and momentum confinement. The isotope effect was studied in detail in the ASDEX tokamak [400], [401] and a good review of the experimental results up to 1992 can be found in [20]. A stronger dependence has been reported in the D - T confinement studies in TFTR [402]. In this case, the energy confinement time scaling is $\tau_E \propto M_i^{0.89 \pm 0.20}$. For fixed local parameters, the scaling of the diffusivities with M_i is even stronger. In TFTR plasmas, the reflectometry

and microwave scattering measurements do not show change within the error bars (20%) on the density fluctuation level between D - D and D - T plasmas. Radial correlation length of turbulence is slightly shorter in D - T plasma, 0.8 cm to be compared with 1.0 cm.

The main issue in understanding the isotope scaling is that simple theoretical scaling gives the opposite dependence on the ion mass. For Bohm scaling—the empirical scaling of the L-mode confinement regime—the diffusivities should be independent of the ion mass. For a pure gyroBohm scaling—the scaling of the theoretical transport models for the core plasma—the diffusivities should scale as $\chi \propto \sqrt{M}$. Naturally, a dependence on the mass ratio is possible in the function F (14) which should be determined by the dynamics of transport. Mechanisms based on atomic physics or edge effects are not effective because the positive scaling with mass has been observed over most of the plasma cross section.

Taking the fluctuation measurement in TFTR at face value, they caution us in using these simple scaling arguments. Again, the need for measuring the turbulent core fluxes for these experiments is clear.

D. Electron Transport

The study of the anomalous electron transport was the focus of the research in the 1970's. Since experiments in the last ten years operate in ion-dominated regimes, not much work has recently been dedicated to the electron-anomalous transport. However, the suppression of anomalous transport in the ion channels has again brought up the problem of understanding electron transport. Some fundamental issues are the following.

- Perturbative transport studies provide mostly information on the electron-channel losses. A recent review by Lopes Cardozo [105] covers most of the relevant issues. Many results from these studies are inconsistent with conventional local-transport models. We have already encountered some of the most remarkable inconsistencies when we discussed the issue of the fast time scales.
- *Magnetic-Induced Losses.* The relative importance of magnetic and electrostatic fluctuations has been a long-standing controversial issue [403]–[405]. The existence of magnetic islands or stochastic magnetic field line regions has been a favorite explanation for the electron anomalous losses. There are two main issues in the theoretical formulation of these type of losses.

— *Magnetic field line-induced diffusivity.* The first evaluations of magnetic-induced losses were based on test particle diffusion in a fully stochastic magnetic field lines [406], [407]. Since the assumption of fully stochastic field lines is not realistic, diffusivities with mixture of stochastic and unbroken surfaces have also been evaluated [408], [409]. The magnetic field-induced particle fluxes are constrained to be ambipolar [410]. That is, the ion-thermal velocity controls the flow of particles. The heat flux based on simple renormalized turbulent response is not ambipolar [411], therefore, the heat flow should be governed by the electron velocity.

However, when electron bunching is taken into account in evaluating the renormalized response, the heat flux becomes ambipolar [412]. This result is supported by experimental measurements of the energy flux in MST, but it is not necessarily the case for tokamaks.

- *Dynamical mechanisms for the magnetic field line perturbation.* This issue is even more complex. Because of the strong tearing mode turbulence at the RFP core, the RFP dynamics is clear. However, the situation is quite different in tokamaks. A variety of magnetic flutter [413] induced by electrostatic instabilities have been considered. The effect on the overall transport was considered to be weak, because the ion-channel losses induced through magnetic flutter are down by $\sqrt{m_e/m_i}$ times the electron losses. Now the light of having better models for the ion channel, some of these models should be revisited. For instance, the current diffusion model has many appealing features [414], [415] and it has been applied to tokamak and stellarator transport. Electromagnetic models range from semiempirical critical gradient model [416] to theoretical models like the magnetic self-organization [417], current filamentation [418], magnetic bubbling [409], and microtearing-induced magnetic turbulence [419].

The experimental situation on magnetic-induced losses remains unclear. In early experiments, some correlation between the magnetic fluctuations and transport was observed [420]. From these experiments no causal relationship can be inferred. The indirect measurements using runaway electrons in ASDEX [109] gave transport levels compatible with energy balance, but again the interpretation is not simple and requires a refinement of the theoretical models for particle diffusion in stochastic magnetic fields. More direct observations are the striations observed in the H_α signal during pellet injection in Tore Supra. They have been interpreted as the interception of rational surfaces by the pellet and thus inferred the existence of magnetic islands [421]. In the same device, radial magnetic field perturbations measured by the cross-polarization scattering diagnostic correlate with confinement and magnetic-island-induced losses seem to dominate confinement in auxiliary heated discharges [422], but the measured magnetic field perturbation is at high k values and the total $\bar{B}_r$ must be inferred from it. Evaluations of the magnetic-field-induced transport in TEXT give a negative result [423]. At the plasma edge and using the same techniques as that in MST [102], it is found that the electron energy flux induced by the magnetic perturbations is too small to account for the whole energy flux [424]. At the plasma core, several techniques have been used to estimate the magnetic fluctuations and infer from them the electron thermal diffusivity [108], [425]. In all cases it was found to be too small to explain the measured electron thermal diffusivity (Fig. 7).

Another source of evidence for magnetic-fluctuation-induced transport is the electron temperature profiles measured by Thomson scattering. The experimental resolution in the measurement of electron temperature profiles has increased,

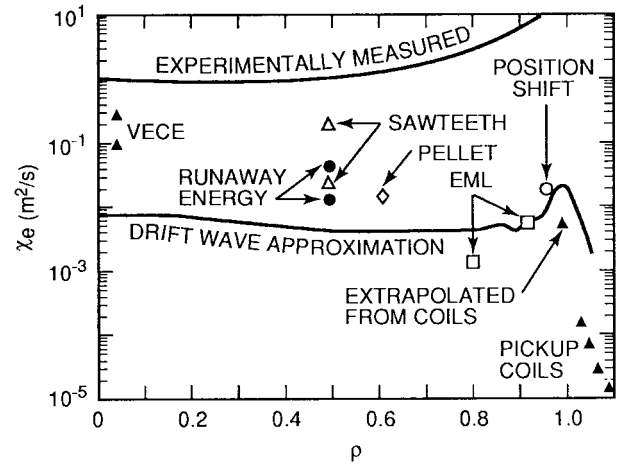


Fig. 7. Magnetic fluctuation-induced transport. Various estimates of electron thermal diffusivity χ_e from experimental inferences of magnetic fluctuation level (data points) and from driftwave theory. The actual χ_e from power balance is shown for comparison [425].

and it is possible to resolve lengths scales on the order of 1 cm. These measurements have brought forward evidence of the presence of structures in the plasma core. The measured electron temperature profiles in JET [426] and TFTR [427] in high- q operation indicate the existence of numerous flat spots, a few centimeters in size, in the profile. These results are reinforced with the high-radial resolution measurements (1.7 mm) in the RTP tokamak showing current filamentation [428]. This evidence has been counter by ECE measurement [429] of the electron temperature profile using a fixed array and moving the plasma in such a way that the full profile can be reconstructed with a resolution of less than 1 cm. No evidence for structures was found, except when a magnetic island was known to exist in the plasma. Recent experimental results from TEXT-U offered an alternative interpretation of the filamentation as double $m = 1$ magnetic islands in the plasma core. This conflicting evidence on the role of magnetic fluctuations in tokamak transport needs serious experimental consideration in the future.

Finally, the different response of the ion and electron channels to the internal transport barrier formation suggest the existence of different transport mechanisms in each channel. Studying the barrier formation can provide insights on the basic mechanism in operation in the electron channel.

E. Plasma-Edge Turbulence Drives

The plasma edge is the best documented region of the plasma in terms of fluctuation measurements [16], [20], [173]. We have empirical evidence that transport in this region is mostly caused by electrostatic turbulence (Section IV). However, we have not yet identified what is the basic turbulence-driving mechanism.

The universal character of the edge turbulence eliminates turbulence models such as the rippling-mode-induced turbulence, which requires $E_{||} \neq 0$. In general, the high level of fluctuations is difficult to explain. Some proposed drives are the following.

- Curvature drives have been proposed as the main turbulence-driving mechanism for tokamaks [430], [431], stellarators [266], [415], and RFP's [265], [268]. The in-out asymmetry of the fluctuations is an argument for curvature drive. However, the theoretical models give fluctuation levels too low to explain the very high levels ($>20\%$) at the edge [179]. In L-mode discharges, the resistive ballooning turbulence has been used in transport modeling of tokamak discharges [286] because of its magnetic shear and q dependence, as a possible mechanism to explain the current scaling of the global confinement. Some of the edge models combine the curvature drive with η_i instability [432], [433].
- Impurity density gradient-driven turbulence [434]. The underlying instability is an equivalent to an η_i mode for impurity species. Therefore, it would be a logical complement to the main ion instability. Experiments in T-10 plasma edge have identified some features that could be consistent with this instability [435]. Detailed theoretical studies should be carried out with the corresponding experimental comparisons.
- Dissipative drift waves have been suggested [246], but the fluctuation levels are too low. A nonlinear mechanism, like the one causing the self-sustainment of drift-wave turbulence [436], [437] could play a role enhancing the fluctuation level.
- Atomic physics drives can lead to high fluctuation levels, and they have been used to reproduce some of the properties of the ohmic edge plasmas. Radiation [438]–[440] and ionization [441], [442] can drive instabilities, and they could be responsible for the edge turbulence. This could provide the link between the modification of edge confinement properties by the control of neutral gas and impurities. This edge fluctuation model is rather complex, and a detailed determination of these atomic physics drives is needed to make it a predictive model. There are some controversial issues in relation to the validity of these drives [443], [444].
- In the SOL—the presence of limiters or the divertor modifies the turbulence model by introducing a different geometry (open field lines), geometrical dependence of the ionization source [445], [446], and/or by introducing a new instability mechanism, the gradient of the electron temperature at the limiter [447], [448].

Experimental measurements have not led to any clear conclusion about those drives. Presence of impurities affect turbulence and transport. The measured electron temperature fluctuation levels comparable to density fluctuation levels and their relative phase close to π favor the radiation drive. However, the injection of impurities changes the overall edge conditions and it is difficult to have a clear conclusion. No direct effect from ionization sources on the level-edge turbulence have been observed and the proximity to a limiter does not seem to influence its asymmetry [449].

The fluctuation levels at the plasma edge in ohmic and L-mode regimes are high ($>50\%$). This raises the question of existence of an equilibrium. The whole concept of plasmas edge, inside and outside the separatrix, needs reexamination.

F. Impact of Plasma-Wall Interactions on Confinement

The effects of plasma-wall interactions on plasma transport and on the achievement of advanced confinement regimes are among the most essential and at the same time least known ingredients to high-performance plasmas. Wall-conditioning methods such as helium-glow discharge conditioning or surface treatment by boronization or gettering with lithium or beryllium have led to the single most effective advancement in confinement. And yet, the underlying physics mechanisms for these phenomena are not known today. There is neither a theoretical model nor experimental evidence that can quantify the effects of wall conditions on plasma performance.

Whereas the equilibrium modeling of the plasma edge, mostly outside the last closed magnetic flux surface, has made a great deal of progress [450]–[453] with detailed incorporation of the magnetic configuration geometry, transport is described by empirical coefficients rather than physics models. In order to build a more complete model of the physics mechanisms of transport in the plasma edge, detailed studies of wall interactions need to be carried out and incorporated in the existing physics models of edge transport, which then need to be combined with the existing fluid descriptions. Bringing these various elements together in the description of the edge plasma is a challenge for theorists, modelers, and experimentalists.

VII. LOOKING TO THE FUTURE

An exciting path is open to the future. Key advances can be expected both in high performance and understanding of transport mechanism. In plasma performance, there are two main challenges ahead.

- Sustainment of the advanced confinement modes toward steady-state operation. In particular for the ERS/NCS confinement modes, sustainment of the q profile for long pulses is a critical issue.
- Control of the transport barrier. If the $E \times B$ shear is the universal mechanism, its control becomes critical for operation in the advanced modes. Flow drive has been suggested [454] as an effective way to provide this control and the CH-mode seems to be a result of this method.

These issues should be addressed in reactor-grade plasmas and the techniques used should extrapolate to a reactor regime. Steady state operation in these confinement modes should be the final goal. High-performance goals should not be limited to tokamaks. Other magnetic concepts should be explored for performance in the advanced regimes. Of particular interest for both performance and understanding is the H-mode in the stellarator. Present results show a limited performance enhancement which should be understood and hopefully improved. The RFP is pursuing successfully the current control path to confinement improvement. It will be interesting to see also if electric field control techniques can be applied to this configuration.

In understanding transport, we have to resolve the open issues listed in the previous section. In many cases, it requires careful experiments with a complete set of transport and fluctuation diagnostics. Because of the role that electric field

shear plays in the enhanced confinement regimes, an electric field diagnostic is particularly needed.

- Scaling studies using dimensionally similar discharges must be done with detailed core fluctuation measurements and with velocity-profile measurements. Theoretical models of transport and fluctuation dynamics must be compared with these results.
- Systematic studies of core fluctuations for the different confinement regimes with radial electric field measurements are critical in understanding the dynamics of the plasma in those regimes. Transition models must be incorporated in full-transport models and tested.
- The role of conditioning on the core transport and, in general, the coupling of the divertor-SOL-edge plasma are issues that require more serious research effort experimentally, theoretically, and computationally.
- Perturbative experiments should be carried out in larger devices with complete diagnostics and used to test our transport understanding. Theoretical models should be developed to interpret the results, existing and future.

Understanding turbulence and transport is a continuing pursuit. It is a fundamental problem in plasma science, and in science in general, due to the ubiquitous nature of turbulence. The increased sophistication in diagnostics and enhanced capabilities for plasma control give to the fusion program an opportunity to make fundamental contributions to this field. It is also a critical issue in fusion research. It is not enough to eliminate anomalous transport at the tokamak core. Reaching to practical applications of fusion involves many technical issues. Looking far into the future and depending on the particular application of the fusion power, different trade-offs may be needed. It is not yet clear that a unique magnetic configuration will be the answer to the various possible applications of fusion. Understanding transport will allow one to optimize the magnetic configuration(s) for these different energy-producing commercial applications.

ACKNOWLEDGMENT

The author would like to thank J. D. Callen, J. W. Connor, P. H. Diamond, W. Dorland, J. N. Leboeuf, V. E. Lynch, K. McGuire, P. K. Mioduszewski, D. Newman, T. Simonen, D. Spong, M. Wakatani, and A. J. Wootton for useful discussions while preparing this paper.

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